



12 PHYSICS

MOTION & FORCES

VECTORS IN TWO DIMENSIONS

Vectors have a size and direction that must be specified at all times.

e.g. displacement, velocity, acceleration, force, momentum, impulse, etc.

Scalars have size only - no direction is necessary.

e.g. distance, speed, work, energy, power, etc.

If two vectors are acting at the same point, it is easiest to use the **parallelogram method** to add them.

Note

Even if the vectors don't act at the same point, this method proves to be the best for solving problems.

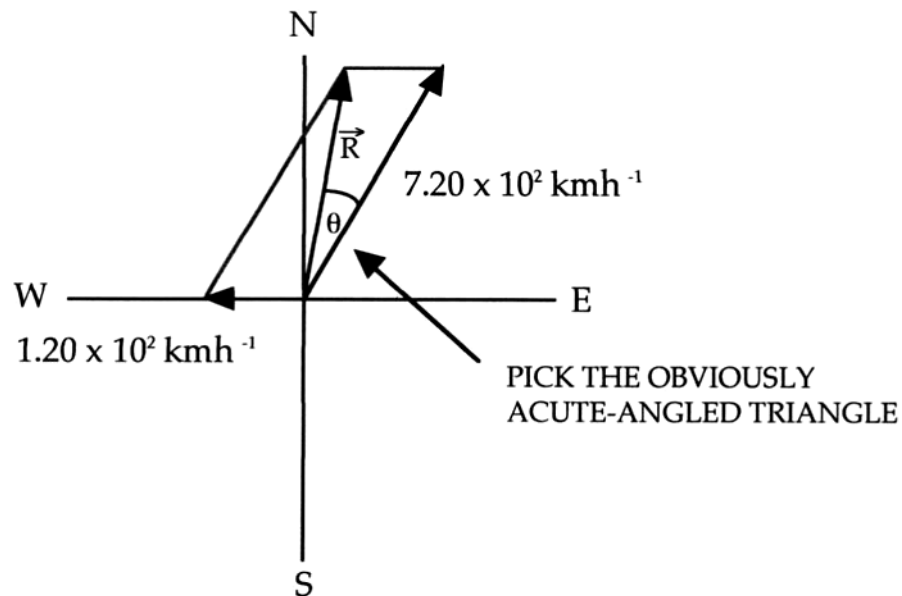
The "rules" for using this method are simple.

1. Draw in a set of "cross-hairs" (with compass directions if appropriate).
2. Draw the two vectors originating at the centre. Make their size **roughly to scale**.
3. Construct the other two sides of the parallelogram.
4. Rule in the resultant vector, which is the **diagonal starting at the centre**.
5. Of the two triangles formed, choose the one with an **obviously acute angle** near the centre of the cross-hairs.

Examples

1. Plane Flying

A plane flying at $7.20 \times 10^2 \text{ kmh}^{-1}$ N30.0°E has to contend with a wind of $1.20 \times 10^2 \text{ kmh}^{-1}$ from the east. Determine the resultant velocity of the plane over the ground.



$$\begin{aligned}\vec{R} &= \sqrt{(7.20 \times 10^2)^2 + (1.20 \times 10^2)^2 - 2(7.20 \times 10^2)(1.20 \times 10^2) \cos 60.0^\circ} \\ &= 6.681 \times 10^2 \text{ kmh}^{-1} \\ &= 1.856 \times 10^2 \text{ ms}^{-1}\end{aligned}$$

$$\begin{aligned}\text{To find } \theta: \quad \frac{6.681 \times 10^2}{\sin 60.0^\circ} &= \frac{1.20 \times 10^2}{\sin \theta} \\ \Rightarrow \theta &= 8.949^\circ\end{aligned}$$

$$\therefore \quad \underline{\vec{R} = 1.86 \times 10^2 \text{ ms}^{-1} \text{ N}21.0^\circ\text{E}}$$

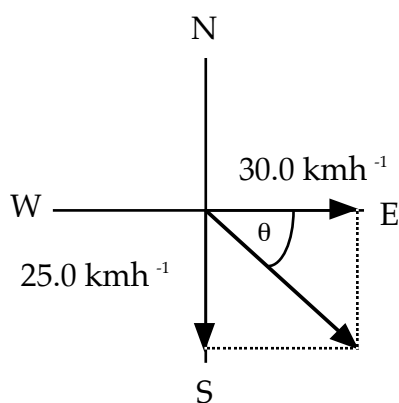
2. Change In Velocity

This is an example of *subtraction of vectors* since $\Delta v = v - u$.

The basic rule is:

“When subtracting a vector, change its direction 180° and add it to the other vector”.

A car travelling at 30.0 kmh^{-1} due west turns left and moves at 25.0 kmh^{-1} . Calculate the change in velocity of the car.



$$\begin{aligned}\Delta v &= v - u \\ &= 25.0 \text{ kmh}^{-1} \text{ S} - 30.0 \text{ kmh}^{-1} \text{ W} \\ &= 25.0 \text{ kmh}^{-1} \text{ S} + 30.0 \text{ kmh}^{-1} \text{ E}\end{aligned}$$

CHANGE DIRECTION 180°,
THEN ADD THE VECTORS.

$$\begin{aligned}R &= \sqrt{(25.0)^2 + (30.0)^2} \\ &= 39.05 \text{ kmh}^{-1} \\ &= 10.85 \text{ ms}^{-1}\end{aligned}$$

$$\begin{aligned}\tan \theta &= \frac{25.0}{30.0} \\ \Rightarrow \theta &= 39.81^\circ\end{aligned}$$

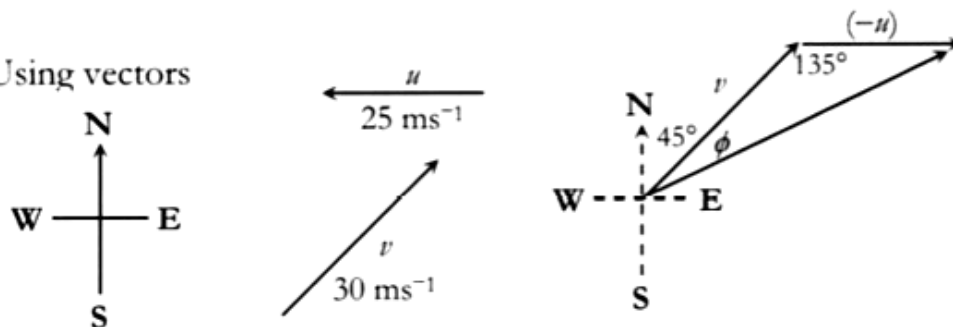
$$\therefore \underline{\Delta v = 10.8 \text{ ms}^{-1} \text{ E}39.8^\circ \text{S}}$$

3.

During a tennis game a boy receives a serve which is travelling at 25 ms^{-1} West just before contact with his racquet. He hits the ball such that it leaves the racquet at 30 ms^{-1} North East. Determine the change in velocity of the ball.

$$\begin{aligned} u &= 25 \text{ ms}^{-1} \text{ W} & \text{Change in velocity } \Delta v &= v - u \\ v &= 30 \text{ ms}^{-1} \text{ NE} & &= v + (-u) \\ av &= ? \end{aligned}$$

Using vectors



Using cosine rule

$$\begin{aligned} R^2 &= (30)^2 + (25)^2 - 2(30)(25) \cos 135^\circ \\ &= 50.9 \text{ ms}^{-1} \end{aligned}$$

Using sine rule

$$\begin{aligned} \frac{R}{\sin 135} &= \frac{\mu}{\sin \phi} \\ \therefore \sin \phi &= \frac{(25)(\sin 135)}{50.9} \end{aligned}$$

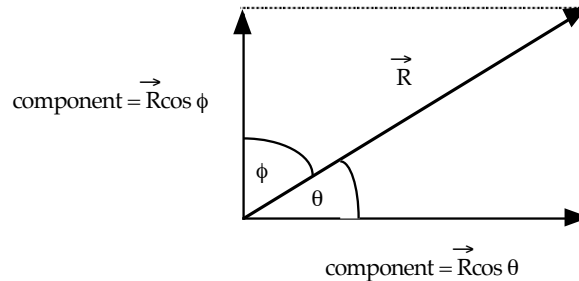
$$\phi = 20.3^\circ$$

$\therefore$ Change in velocity of the ball = 50.9 ms^{-1} $65.3^\circ \text{ E of N}$.

COMPONENTS OF A VECTOR

Any single vector can be expressed as *two component vectors that are at right angles to each other*.

Neither component vector can be bigger than the original vector.



Any component is given by:

$$\text{component} = R \cos \theta$$

where R = given vector.

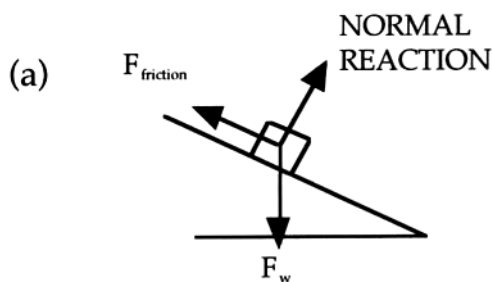
θ = the angle between R and the direction of the required component.

Examples

1. Inclined Plane

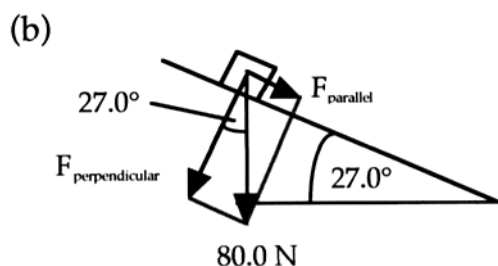
A box weighing 80.0 N sits at rest on a plane inclined at 27.0° to the horizontal.

- (a) Draw a diagram showing all forces acting on the box.
- (b) Calculate the force exerted perpendicular to the plane by the box onto the plane.



Only three forces act on the box.

The components of F_w are **not** forces that are acting.



$$\begin{aligned} F_{\text{perp.}} &= F_w \cos 27.0^\circ \\ &= 80.0 \cos 27.0^\circ \\ &= 71.28 \text{ N} \end{aligned}$$

$$\therefore F_{\text{perp.}} = 71.3 \text{ N into the plane.}$$

2.

A box of mass 12.5 kg is sliding down a frictionless plane, inclined at 35° to the horizontal. Determine the force.

- (a) (i) acting down the plane
(ii) into the plane

(b) Assuming $g = 9.80 \text{ ms}^{-2}$, determine the acceleration of the mass down the plane.

- (a) Force $\parallel$ to the plane

$$\begin{aligned} F_{\parallel} &= m g \sin \theta \\ &= (12.5)(9.80)(\sin 35^\circ) \\ &= 70.3 \text{ N} \end{aligned}$$

Force perpendicular to the plane

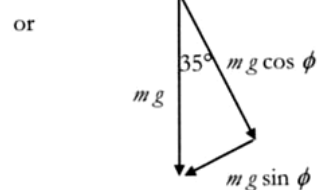
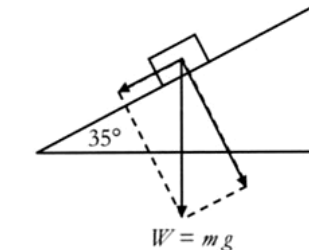
$$\begin{aligned} F_{\text{perp}} &= m g \cos \theta \\ &= (12.5)(9.80)(\cos 35^\circ) \\ &= 100 \text{ N} \end{aligned}$$

- (b) Since

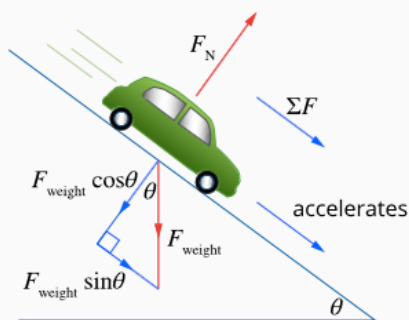
$$\begin{aligned} F &= m a \\ a &= \frac{F}{m} \end{aligned}$$

$$\parallel \text{ to plane } a = \frac{m g \sin \theta}{m}$$

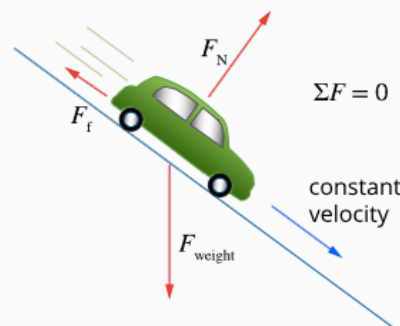
$$\begin{aligned} a &= (9.8) (\sin 35) \\ &= 5.62 \text{ ms}^{-2} \end{aligned}$$



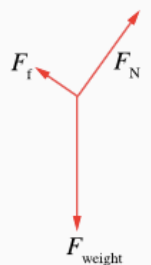
3. Consider the forces acting on a smooth incline (no friction) and one with friction.



2.1.4 • The motion of an object on a smooth inclined plane can be analysed by finding the components of the weight force that are parallel and perpendicular to the plane.



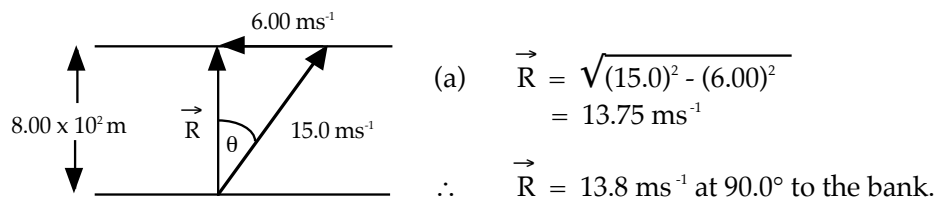
2.1.5 • If friction acts on the car, causing it to move with a constant velocity, the net force on the car will be zero, and so the forces will be in balance both perpendicular and parallel to the incline.



4. River Problem

A person wants to cross a river $8.00 \times 10^2 \text{ m}$ wide to reach a jetty directly across on the opposite bank. If the river is flowing at 6.00 ms^{-1} and the boat that is used can travel at 15.0 ms^{-1} in still water, calculate:

- the resultant velocity of the boat in relation to the river bottom (or bank).
- the time taken to reach the other side.



- To calculate the time taken, use a component of any velocity that is directly across the river.

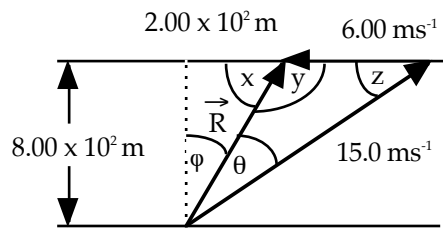
In this example, $\vec{R}$ is directly across so use that.

$$\text{i.e. } v_{\text{across}} = \frac{s_{\text{across}}}{t_{\text{across}}}$$

$$\begin{aligned} \Rightarrow t &= s / v \\ &= 8.00 \times 10^2 / 13.75 \\ &= 58.18 \text{ s} \end{aligned}$$

$$\therefore \text{Time taken} = 58.2 \text{ s}$$

5. If the person in Question 3 was trying to reach a jetty $2.00 \times 10^2 \text{ m}$ upstream, calculate the resultant velocity and the time taken.



$$\tan \phi = 2.00 \times 10^2 / 8.00 \times 10^2$$

$$\Rightarrow \phi = 14.04^\circ$$

By geometry:

$$x = 75.96^\circ$$

$$y = 104.04^\circ$$

To find θ :

$$\frac{6.00}{\sin \theta} = \frac{15.0}{\sin 104.04^\circ}$$

$$\Rightarrow \theta = 22.83^\circ$$

By geometry:

$$z = 53.13^\circ$$

To find R :

$$\frac{\vec{R}}{\sin 53.13^\circ} = \frac{6.00}{\sin 22.83^\circ}$$

$$\Rightarrow \vec{R} = 12.37 \text{ ms}^{-1}$$

$\therefore \vec{R} = 12.4 \text{ ms}^{-1}$ at 53.1° to the bank upstream.

To calculate the time taken to cross, use any component of velocity that is directly across the river.

e.g. $12.37 \cos 14.04^\circ$ or $15.0 \cos 36.87^\circ$

$$v_{\text{across}} = \frac{s_{\text{across}}}{t_{\text{across}}}$$

$$\Rightarrow t = s / v$$

$$= 8.00 \times 10^2 / 12.37 \cos 14.04^\circ$$

$$= 66.66 \text{ s}$$

$\therefore$ Time taken = 66.7 s .

ADDING MORE THAN TWO VECTORS

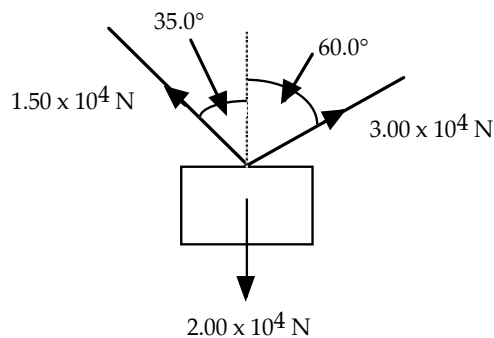
The best way to solve these problems is to use **components** to reduce the problem to the addition of two vectors at right angles to each other.

Generally this will be done by considering movement in two dimensions within a plane.

- i.e. up / down / left / right
north / south / east / west (if compass directions are given).

Example

A heavy crate is being lifted by two cables as shown below.



If the crate weighs $2.00 \times 10^4 \text{ N}$ and the tension in each cable is as shown, calculate the resultant force experienced by the crate.

VERTICALLY

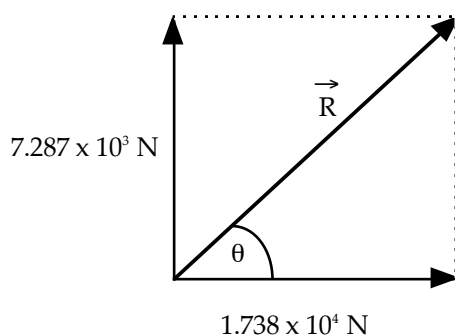
Take up as +ve.

$$\begin{aligned} F_v &= 1.50 \times 10^4 \cos 35.0^\circ + 3.00 \times 10^4 \cos 60.0^\circ - 2.00 \times 10^4 \\ &= 7.287 \times 10^3 \text{ N} \end{aligned}$$

HORIZONTALLY

Take to the right as +ve.

$$\begin{aligned} F_h &= 3.00 \times 10^4 \cos 30.0^\circ - 1.50 \times 10^4 \cos 55.0^\circ \\ &= 1.738 \times 10^4 \text{ N} \end{aligned}$$



$$\begin{aligned} \vec{R} &= \sqrt{(7.287 \times 10^3)^2 + (1.738 \times 10^4)^2} \\ &= 1.885 \times 10^4 \text{ N} \end{aligned}$$

$$\begin{aligned} \tan \theta &= 7.287 \times 10^3 / 1.738 \times 10^4 \\ \Rightarrow \theta &= 22.75^\circ \end{aligned}$$

$$\therefore \underline{\underline{\vec{R} = 1.88 \times 10^4 \text{ N at } 22.8^\circ \text{ above the horizontal and to the right.}}}$$

EQUATIONS OF MOTION

From Year 11, the equations of motion are:

$$v = u + at$$

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

These equations only apply to *accelerated motion in one dimension*.

Generally this was applied to:

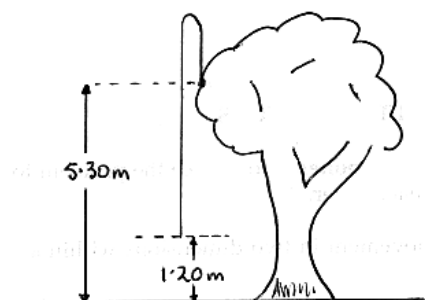
- (a) objects accelerating/decelerating across the ground, or
- (b) objects moving up/down under the influence of gravity.

Revise your notes on this section from last year .

Example

A ball is thrown upwards from a point 1.20 m above the ground and becomes stuck in a tree branch 5.30 m above the ground on the way down from its highest point. If the time of flight was 3.25 s, calculate:

- (a) the initial velocity of the ball.
- (b) the maximum height the ball reached.
- (c) the velocity of the ball at impact with the branch.



Take down as +ve.

$$\begin{aligned}
 v &= ? \\
 u &= ? \\
 a &= 9.80 \text{ ms}^{-2} \\
 t &= 3.25 \text{ s} \\
 s &= -4.10 \text{ m}
 \end{aligned}
 \quad
 \begin{aligned}
 s &= ut + \frac{1}{2}at^2 \\
 \Rightarrow -4.10 &= u(3.25) + \frac{1}{2}(9.8)(3.25)^2 \\
 \Rightarrow u &= -17.19 \text{ ms}^{-1} \\
 \therefore u &= 17.2 \text{ ms}^{-1} \text{ upwards.}
 \end{aligned}$$

At the highest point, the ball will stop.

↓ +ve

(b) $v = 0.00 \text{ ms}^{-1}$ $u = -17.19 \text{ ms}^{-1}$ $a = 9.80 \text{ ms}^{-2}$ $t = ?$ $s = ?$	$v^2 = u^2 + 2as$ $\Rightarrow 0.00 = (-17.19)^2 + 2(9.8)s$ $\Rightarrow s = -15.08 \text{ m}$ $\therefore$ <u>Max. height = 16.3 m above the ground.</u>
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↓ +ve

(c)	$v = ?$	$v = u + at$
	$u = -17.19 \text{ ms}^{-1}$	$= -17.19 + (9.80)(3.25)$
	$a = 9.80 \text{ ms}^{-2}$	$= 14.66 \text{ ms}^{-1}$
	$t = 3.25 \text{ s}$	
	$s = -4.10 \text{ m}$	$\therefore \underline{v = 14.7 \text{ ms}^{-1} \text{ downwards.}}$

For an object not undergoing acceleration but rather is moving at constant velocity, the equation applied is:

$$\boxed{v = \frac{s}{t}}$$

All of these equations of motion are utilised in analysing *projectile motion*.

PROJECTILE MOTION

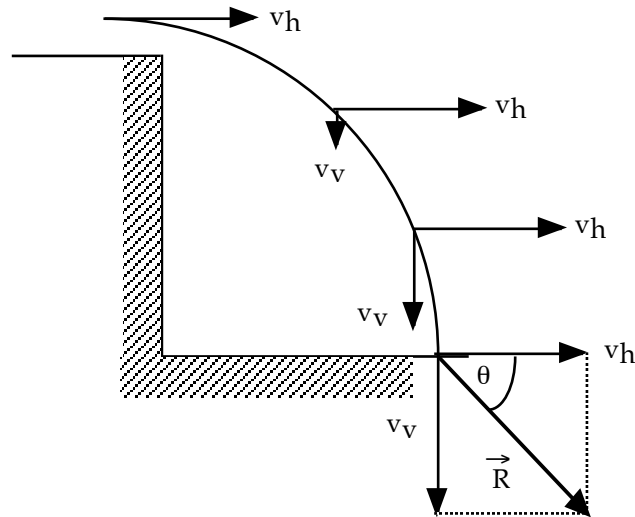
For an object thrown horizontally, or at an angle to the horizontal, in the Earth's gravitational field, its ***vertical and horizontal motions are independent of one another.***

Horizontally

- Friction due to air resistance is ignored.
- Horizontal velocity (v_h) remains ***constant***.
- ***Must use $v = \frac{s}{t}$ for calculations.***

Vertically

- Friction due to air resistance is ignored.
- Objects accelerate downwards due to gravity.
- ***Must use the three equations of motion for acceleration in all calculations.***

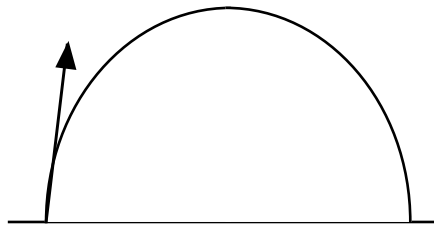


At impact with the ground, the ***resultant velocity is the vector addition of the vertical and horizontal velocities.***

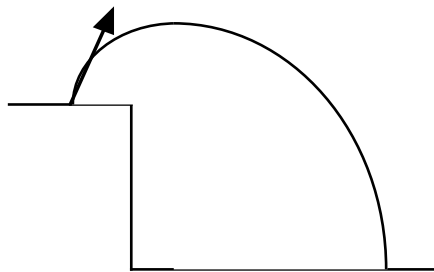
An ***angle θ to the ground*** should also be calculated as the impact velocity is a vector.

There are several different types of projectile motions possible (apart from the one shown above).

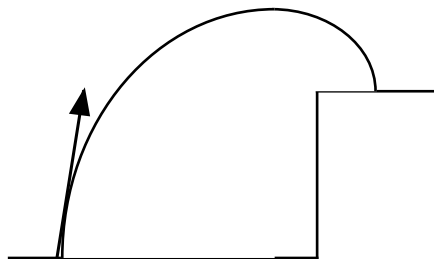
1. Object projected upwards at an angle and caught at the same level.



2. Object projected upwards at an angle and finishing below its release height.



3. Object projected upwards at an angle and finishing above its release height.



▲ Tips for problems involving projectile motion



- Construct a diagram showing the motion and set the problem out clearly. Show the information related for each component, $\updownarrow$ vertical and $\leftrightarrow$ horizontal, separately.
- In the $\leftrightarrow$ horizontal component, the velocity, v , of the projectile is constant, so the only formula needed is $v_{av} = \frac{s}{t}$.
- For the $\updownarrow$ vertical component, the projectile is moving with a constant acceleration (9.8 m s^{-2} down), so the equations of motion for uniform acceleration must be used:
$$v = u + at$$
$$s = ut + \frac{1}{2}at^2$$
$$v^2 = u^2 + 2as$$
- In the $\updownarrow$ vertical component, it is important to clearly specify whether up or down is the positive or negative direction, and use this consistently throughout the problem.
- If a projectile is launched horizontally, its initial speed is the same as the horizontal component throughout the flight.
- If a projectile is launched at an angle to the horizontal, trigonometry can be used to find the initial horizontal and vertical speeds.

- Pythagoras' theorem can be used to determine the actual speed of the projectile.
- If the velocity of the projectile is required, it is necessary to give a direction with respect to the horizontal as well as the speed of the projectile.
- It is also possible to solve projectile problems using equations for kinetic energy and gravitational potential energy, as discussed in 'The force due to gravity' chapter. These include:

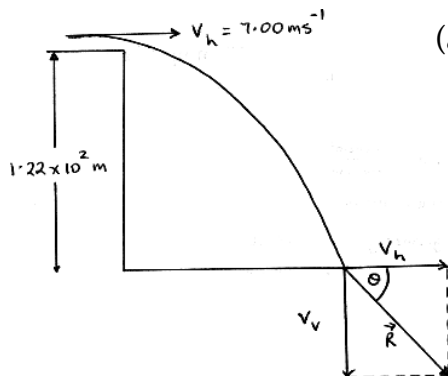
$$E_k = \frac{1}{2}mv^2$$

$$E_p = mgh$$

Examples

1. A stone is thrown horizontally off a cliff at 7.00 ms^{-1} . If the cliff is $1.22 \times 10^2 \text{ m}$ high, calculate:

- the time taken to reach the ground.
- the distance from the base of the cliff the stone hits the ground.
- the impact velocity of the stone.



(a) VERTICALLY

↓ +ve

$$v = ?$$

$$u = 0.00 \text{ ms}^{-1}$$

$$a = 9.80 \text{ ms}^{-2}$$

$$t = ?$$

$$s = 1.22 \times 10^2 \text{ m}$$

$$s = ut + \frac{1}{2}at^2$$

$$\Rightarrow 1.22 \times 10^2 = 0 + \frac{1}{2}(9.80)t^2$$

$$\Rightarrow t = 4.990 \text{ s}$$

$$\therefore \text{Time taken} = 4.99 \text{ s}$$

(b) HORizontally

$$v_h = \frac{s_h}{t}$$

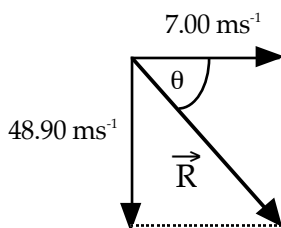
$$\Rightarrow 7.00 = \frac{s_h}{4.990}$$

$$\Rightarrow s_h = 34.93 \text{ m}$$

$\therefore$ Stone hits 34.9 m from the base of the cliff

(c) To calculate v_v :

$$\begin{aligned} v &= u + at \\ &= 0 + (9.80)(4.990) \\ &= 48.90 \text{ ms}^{-1} \end{aligned}$$



$$\begin{aligned} \vec{R} &= \sqrt{(48.90)^2 + (7.00)^2} \\ &= 49.40 \text{ ms}^{-1} \end{aligned}$$

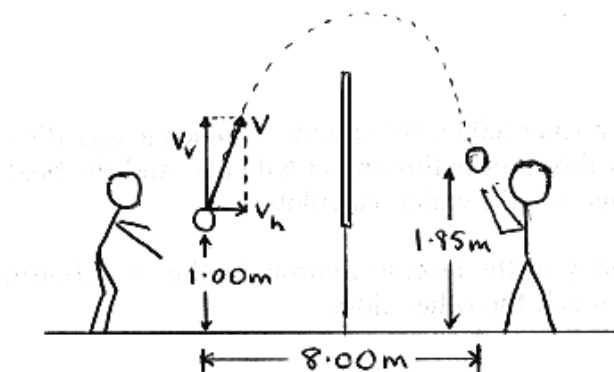
$$\tan \theta = \frac{48.90}{7.00}$$

$$\Rightarrow \theta = 81.85^\circ$$

$\therefore$ Impact velocity = 49.4 ms^{-1} at 81.8° to the horizontal.

2. A volleyball player is 5.40 m from the net when she “digs” the ball 1.00 m from the ground, causing it to rise to a height of 6.20 m above the court. Given that the opposition player on the other side of the net is 2.60 m away from the net when she “sets” the ball, and the ball is 1.85 m above the ground, calculate:

- the vertical component of the initial velocity of the ball.
- the time of flight of the ball.
- the initial velocity of the ball after being struck.



- (a) VERTICALLY

Consider the motion to the top.

$$v = 0.00 \text{ ms}^{-1}$$

$$u = ?$$

$$a = 9.80 \text{ ms}^{-2}$$

$$t = ?$$

$$s = -5.20 \text{ m}$$

$\Rightarrow$

$\Rightarrow$

$\therefore$

$$\downarrow +ve$$

$$v^2 = u^2 + 2as$$

$$0 = u^2 + 2(9.80)(-5.20)$$

$$u = -10.10 \text{ ms}^{-1}$$

$$\therefore \underline{v_v = 10.1 \text{ ms}^{-1} \text{ upwards.}}$$

- (b) VERTICALLY

Consider the whole motion.

$$v = ?$$

$$u = -10.10 \text{ ms}^{-1}$$

$$a = 9.80 \text{ ms}^{-2}$$

$$t = ?$$

$$s = -0.85 \text{ m}$$

$\Rightarrow$

$\Rightarrow$

$\Rightarrow$

$$\downarrow +ve$$

$$s = ut + \frac{1}{2}at^2$$

$$-0.85 = -10.10t + \frac{1}{2}(9.80)t^2$$

$$4.90t^2 - 10.10t + 0.85 = 0$$

$$t = \frac{10.10 \pm \sqrt{(-10.10)^2 - 4(4.90)(0.85)}}{2(4.90)}$$

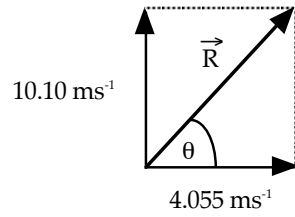
$$= 1.973 \text{ s or } 8.791 \times 10^{-2} \text{ s}$$

$\therefore$

$$\underline{\text{Time of flight} = 1.97 \text{ s}}$$

(c) HORIZONTALLY

$$\begin{aligned}
 v_h &= \frac{s_b}{t} \\
 &= \frac{8.00}{1.973} \\
 &= 4.055 \text{ ms}^{-1}
 \end{aligned}$$

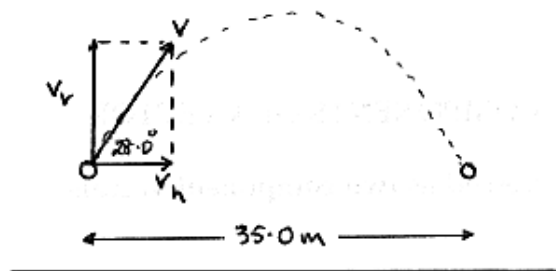


$$\begin{aligned}
 \vec{R} &= \sqrt{(10.10)^2 + (4.055)^2} \\
 &= 10.88 \text{ ms}^{-1}
 \end{aligned}$$

$$\begin{aligned}
 \tan \theta &= \frac{10.10}{4.055} \\
 \Rightarrow \theta &= 68.13^\circ
 \end{aligned}$$

$\therefore$ Initial velocity = 10.9 ms⁻¹ at 68.1° to the horizontal.

3. Two boys are playing “catch” with a cricket ball. They are standing 35.0 m apart when one throws the ball at an angle of 28.0° to the horizontal. At what speed must he throw the ball if the other boy is to catch it at the same height as its release?



VERTICALLY

Consider the whole motion.

$$\begin{aligned}
 v &= ? & s &= ut + \frac{1}{2}at^2 \\
 u &= -v_v & \Rightarrow 0 &= -v_v t + \frac{1}{2}(9.80)t^2 \\
 a &= 9.80 \text{ ms}^{-2} & \Rightarrow v_v &= 4.90 t \\
 t &= ? & \Rightarrow v \cos 62.0^\circ &= 4.90 t \quad \text{Eqn 1} \\
 s &= 0.00 \text{ m}
 \end{aligned}$$

HORizontally

$$\begin{aligned}
 v_h &= \frac{s_h}{t} \\
 \Rightarrow v \cos 28.0^\circ &= \frac{35.0}{t} \\
 \Rightarrow t &= \frac{35.0}{v \cos 28.0^\circ} \quad \text{Eqn 2}
 \end{aligned}$$

Substitute Eqn 2 into Eqn 1.

$$\begin{aligned}
 \Rightarrow v \cos 62.0^\circ &= \frac{(4.90)(35.0)}{v \cos 28.0^\circ} \\
 \Rightarrow v^2 (\cos 28.0^\circ)(\cos 62.0^\circ) &= (4.90)(35.0) \\
 \Rightarrow v &= 20.34 \text{ ms}^{-1}
 \end{aligned}$$

$\therefore$ Release velocity = 20.3 ms^{-1} at 28.0° to the horizontal.

4.

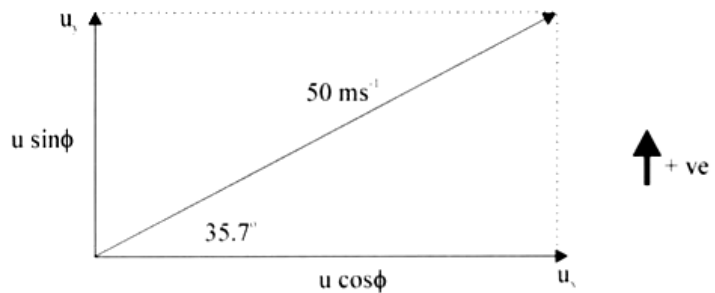
An arrow is fired into the air at 50.0ms^{-1} at an angle 35.7° to the horizontal. Assuming no effects from air resistance determine:

- (a) its maximum height
- (b) its time of flight
- (c) its range.

Determine horizontal and vertical components of initial velocity.

$$\begin{aligned}\text{Horizontal } u_x &= u \cos \phi \\ &= (50)(\cos 35.7) \\ &= 40.6 \text{ ms}^{-1}\end{aligned}$$

$$\begin{aligned}\text{Vertical } u_y &= (50)(\sin 35.7) \\ &= 29.2 \text{ ms}^{-1}\end{aligned}$$



- (a) Now - considering vertical motion only

$$\begin{aligned}u &= 29.2 \text{ ms}^{-1} & v^2 &= u^2 + 2as \\ a &= -9.80 \text{ ms}^{-2} & s &= \frac{v^2 - u^2}{2a} \\ s &= ? & &= \frac{0 - (29.2)^2}{(2)(-9.80)}\end{aligned}$$

$$\text{Maximum height} = 43.5 \text{ m}$$

- (b) Consider vertical motion only - the whole flight

$$\begin{aligned}u &= 29.2 \text{ ms}^{-1} & s &= ut + \frac{1}{2}at^2 \\ a &= -9.80 \text{ ms}^{-2} & 0 &= (29.2)(t) + (0.5)(-9.8)(t^2) \\ s &= 0 & 0 &= 29.2t - 4.9t^2 \\ t &= ? & 0 &= 29.2 - 4.9t \\ & & t &= 5.96 \text{ s} = \text{flight time}\end{aligned}$$

- (c) Consider horizontal motion only. Assume constant velocity:

$$\begin{aligned}v &= 40.6 \text{ ms}^{-1} & s &= vt \\ t &= 5.96 \text{ s} & &= (40.6)(5.96) \\ s &= ? & \text{Range} &= 242 \text{ m}\end{aligned}$$

AIR RESISTANCE

Air resistance affect projectiles in a variety of ways.

The path of an object is not truly parabolic but is asymmetric with a gradual rise and a steeper descent.

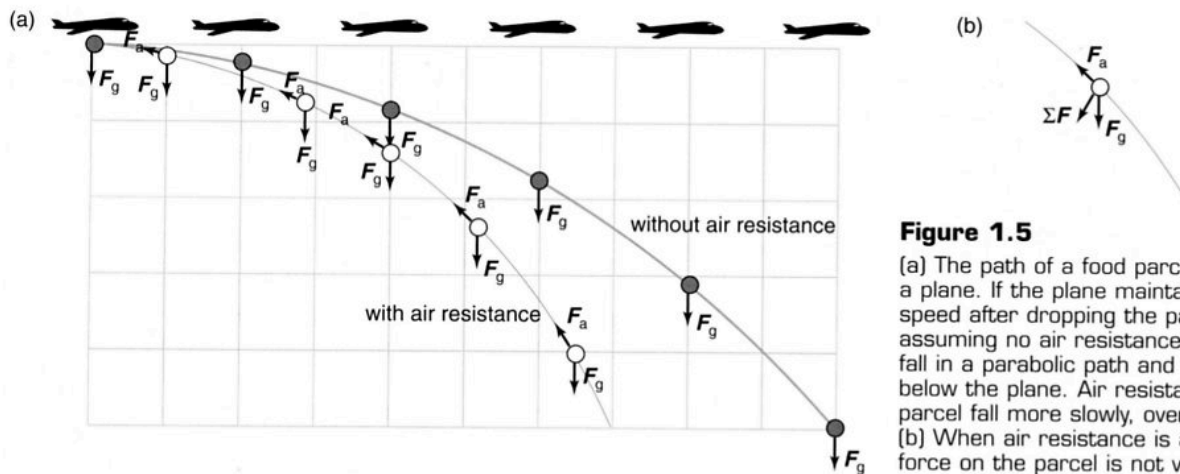
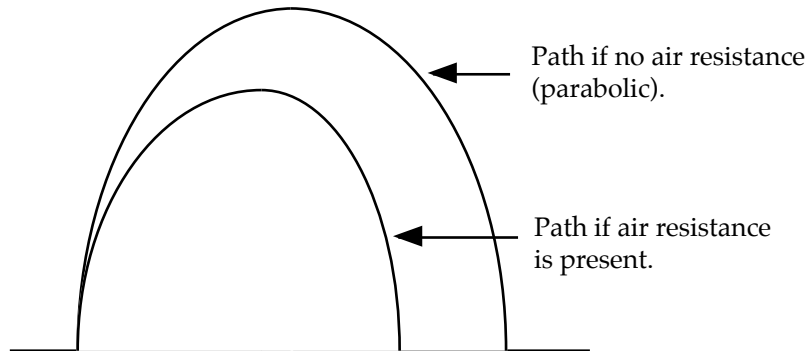


Figure 1.5

(a) The path of a food parcel dropped from a plane. If the plane maintains a constant speed after dropping the parcel and assuming no air resistance, the parcel will fall in a parabolic path and remain directly below the plane. Air resistance makes the parcel fall more slowly, over a shorter path. (b) When air resistance is acting, the net force on the parcel is not vertically down.

Objects that fall vertically through the atmosphere will ultimately reach a constant speed called **terminal velocity**.

At this speed, the **force due to gravity is equalled by the frictional forces** acting upwards, resulting in **no nett force**. Hence there is **no nett acceleration** and the object moves at **constant velocity**.

For most objects, this velocity is about 195 kmh^{-1} .

The actual size of the terminal velocity is determined by several factors.

1. ***Silhouette area*** - the area exposed to the air as the object falls.

A large area will mean a large force of friction.

e.g. Skydivers will open out their arms and legs to increase their silhouette area. This allows them to control their movements in the air and they will have a lower terminal velocity.

Javelins, racing cars, etc are streamlined (giving a small silhouette) and have much smaller air resistance.

2. ***Silhouette area to mass ratio***.

If this ratio is small, then the terminal velocity will be high.

e.g. A shot put has a large mass but small silhouette, so the ratio is small. Hence air resistance has little effect.

A table tennis ball is very light so its ratio is much higher than the shot. Hence it has a smaller terminal velocity and is severely affected by air resistance.

3. ***Spin*** - causes objects to change direction in the air.

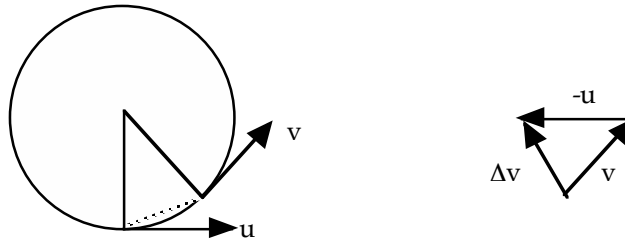
e.g. Balls that spin (tennis, cricket, table tennis, golf, etc) have pressure differences generated on opposite sides of them. As the high air pressure moves towards the low pressure, it moves the ball.

CIRCULAR MOTION

An object moving in a circle at constant speed is always changing the direction of its instantaneous velocity. Thus it must be accelerating.

The direction of this acceleration is ***always towards the centre of the circle.***

This can be shown as follows.



Calculation of Δv for the object moving around the circle is made by adding the vectors $-u$ and v together since:

$$\Delta v = v - u$$

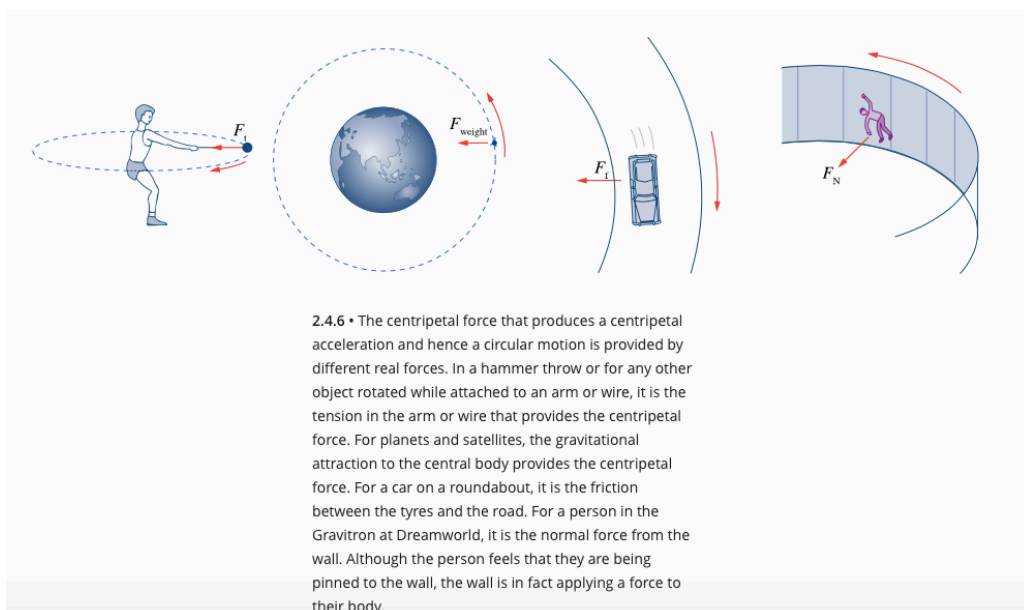
The complete derivation is of no concern here, but it is sufficient to say that the ***direction of Δv is towards the centre of the circle.***

Given that $a = \frac{\Delta v}{t}$, then the acceleration must also be towards the centre of the circle.

From $F = ma$, there ***must be a force in the direction of the nett acceleration.*** (This comes from Newton's Second Law.)

This force is called a ***centripetal force.***

The centripetal force is required to keep the object moving in a circle and can be provided by a ***string*** (or something similar) attached to the object, or ***gravity*** in other situations. All of these will be covered in notes.



The centripetal force is given by:

$$F_c = \frac{mv^2}{r}$$

where F_c = the centripetal force (N)
 m = the mass of the object (kg)
 v = orbital speed of the object (ms^{-1})
 r = the radius of the orbit (m).

The centripetal acceleration on an object is given by:

$$a_c = \frac{v^2}{r}$$

where a_c = centripetal acceleration (ms^{-2}).

The orbital speed of an object is calculated using:

$$v = \frac{2\pi r}{T}$$

where T = the period of revolution (or the time taken for one complete revolution) (s).

Examples

1.

A compact disk is rotating at a rate of 8.00 revolutions per second. Calculate the speed of a point on the inner track of the CD which is moving in a circular path of radius 2.25 cm.

Solution

If the disk is rotating at 8.00 revolutions per second, its period is:

$$\begin{aligned} f &= 8.00 \text{ s}^{-1} & T &= \frac{1}{f} = \frac{1}{8.00} \\ & & &= 0.125 \text{ s} \end{aligned}$$

The speed of a point on the inner track is:

$$\begin{aligned} r &= 2.25 \times 10^{-2} \text{ m} & v &= \frac{2\pi r}{T} = \frac{2\pi(2.25 \times 10^{-2})}{0.125} \\ & & &= 1.13 \text{ m s}^{-1} \end{aligned}$$

At the outer rim of the disk, it has slowed to around 200 rpm or 3.30 revolutions per second. This is done to ensure that, as the disk is played, the laser beam can be drawn along the track at a constant rate.

2.

Young Rachael is sitting on a horse on a merry-go-round which makes 4.2 revolutions per minute. The horse is 3.50 m from the centre. Rachael's mass is 25 kg.

- (a) What is Rachael's period of rotation?
 (b) What is her speed?
 (c) What is the net force exerted on Rachael? How is it applied?

$$f = 4.2 \text{ rpm}$$

$$= ? \text{ Hz}$$

$$r = 3.50 \text{ m}$$

$$m = 25 \text{ kg}$$

$$T = ?$$

$$v = ?$$

$$F_c = ?$$

$$(a) \quad f = \frac{4.2}{60} = 0.07 \text{ Hz}$$

$$\therefore T = \frac{1}{f} = \frac{1}{0.07}$$

$$= 14.3 \text{ s}$$

$$(b) \quad v = \frac{2\pi r}{T} = \frac{(2\pi)(3.50)}{14.3}$$

$$= 1.54 \text{ ms}^{-1}$$

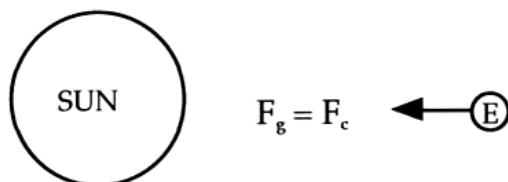
$$(c) \quad F_c = \frac{mv^2}{r} = \frac{(25)(1.54)^2}{3.5}$$

$$= 16.9 \text{ N}$$

This force is applied by the friction between her and the horse.

3. The Earth maintains an approximately circular path around the Sun. The average radius of this orbit is $1.49 \times 10^8 \text{ km}$ and the period is 365.26 days. The Earth's mass is $5.98 \times 10^{24} \text{ kg}$.

What is the average gravitational force of the Sun on the Earth required to maintain this orbit?



$$F_g = F_c$$

$$= \frac{mv^2}{r}$$

$$= \frac{4\pi^2 mr}{T^2} \quad \left[\text{since } v = \frac{2\pi r}{T} \right]$$

$$= \frac{4\pi^2 (5.98 \times 10^{24}) (1.49 \times 10^{11})}{[(365.26)(24.0)(60.0)(60.0)]^2}$$

$$= 3.532 \times 10^{22} \text{ N}$$

$$\therefore \underline{F_{\text{gravity}} = 3.53 \times 10^{22} \text{ N towards the Sun.}}$$

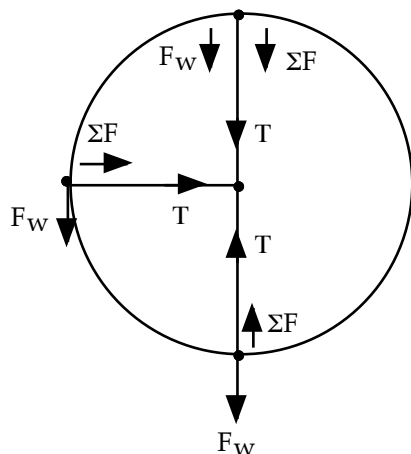
MOTION IN A VERTICAL CIRCLE

In solving problems involving vertical circular motion, always remember that:

“ ΣF is always in the direction of the nett acceleration, which for circular motion is always towards the centre of the circle”.

Use this information to draw clear diagrams to show the forces acting in the situation.

Objects Attached To A String



AT TOP: $\Sigma F = F_c = F_w + T$

AT SIDE: $\Sigma F = F_c = T$

AT BOTTOM: $\Sigma F = F_c = T - F_w$

Note

From last year's work on Newton's Third Law:

“ ΣF = All forces in the same direction as the nett acceleration - all forces in the opposite direction to the nett acceleration”.

Applying this to the situation at the bottom of the circle:

T is in the direction of the nett acceleration (i.e. towards the centre of the circle).

F_w is in the opposite direction to the nett acceleration.

$$\therefore \Delta F = T - F_w$$

Important

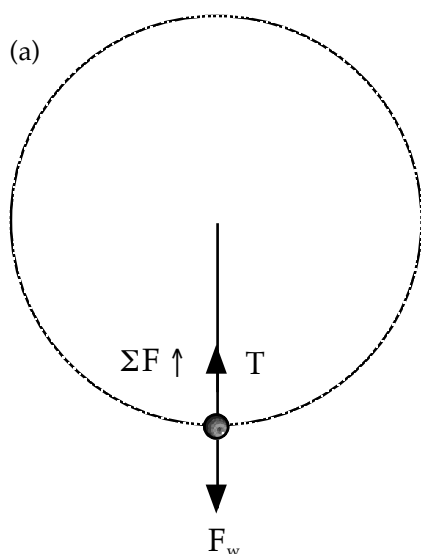
Don't try to learn the above “equations” rotely.

Learn how to draw the diagram and apply Newton's Third Law to derive them. Then you will be able to handle any question or problem presented to you!

Example 1

In an experiment to determine the force required to break a light string, a group of students analysed the motion of a small 30.0 g mass attached to a 75.0 cm piece of the string being whilst being whirled in a vertical circle. One student increased the speed of the mass in the circle until it caused the string to break. The students used a high-speed video to determine that the speed at the breaking point was 13.6 ms^{-1} .

- (a) At which point in the circle would the string most likely have broken? Explain why.
- (b) What is the force required to break the string?



At the bottom of the swing the tension created is the greatest.

$$\begin{aligned} \text{i.e. } \Sigma F &= T - F_w \\ \Rightarrow T &= \Sigma F + F_w \end{aligned}$$

The tension is greater than the weight by an amount equal to the centripetal force due to its motion in a circle.

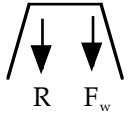
$$\begin{aligned} \text{(b)} \quad \Sigma F &= T - F_w \\ \Rightarrow T &= \Sigma F + F_w \\ &= \frac{mv^2}{r} + mg \\ &= \frac{(0.0300)(13.6)^2}{(0.750)} + (0.0300)(9.80) \\ &= 7.692 \text{ N} \end{aligned}$$

$\therefore$ Force required = 7.69 N.

Example 2

A student whirled a bucket of water above her head at a speed that just allowed the water to stay in the bucket as it passed overhead.

- (a) Explain how it is possible to have the water stay in the bucket.
(b) Given the distance from the student's shoulder to the centre of mass of the bucket and water is 1.10 m, calculate the minimum speed required to keep the water in the bucket.

(a) $\Sigma F \downarrow$  The bucket is swung fast enough to experience a centripetal acceleration downwards that is greater than 9.80 ms^{-2} .

$\Sigma F = F_c = F_w + R$ Hence the bottom of the bucket exerts a force onto the water.

(b) $\Sigma F = F_c = F_w + R$

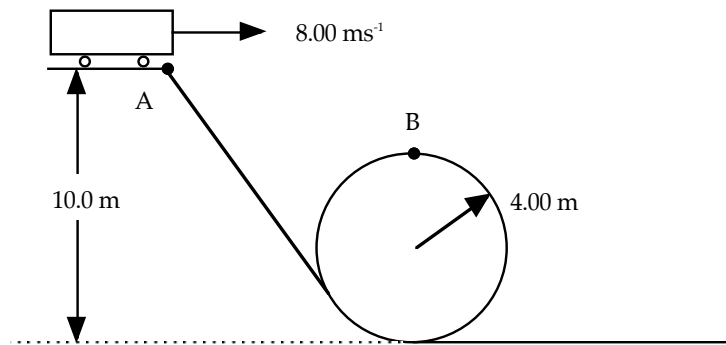
When $R = 0$, the water will just stay inside the bucket.

$$\begin{aligned} \Rightarrow F_c &= F_w \\ \Rightarrow \frac{mv^2}{r} &= mg \\ \Rightarrow v &= \sqrt{gr} \\ &= \sqrt{(9.80)(1.10)} \\ &= 3.283 \text{ ms}^{-1} \end{aligned}$$

$\therefore$ Minimum speed = 3.28 ms^{-1} .

Example 3 (Conservation Of Mechanical Energy)

A roller coaster of mass $2.25 \times 10^3 \text{ kg}$ is moving at 8.00 ms^{-1} at point A in the diagram below.



- Calculate the total energy of the roller coaster at point A.
- Determine whether the coaster is able to safely make it through point B.
- If it does make it through B, calculate the force exerted by the track onto the roller coaster.

(a) **AT A:**

$$\begin{aligned}
 E_t &= E_p + E_k \\
 &= mgh + \frac{1}{2}mv^2 \\
 &= (2.25 \times 10^3)(9.80)(10.0) + \frac{1}{2}(2.25 \times 10^3)(8.00)^2 \\
 &= 2.925 \times 10^5 \text{ J} \\
 \therefore \quad \underline{\text{Total energy} = 2.95 \times 10^5 \text{ J}}
 \end{aligned}$$

(b) **AT B:**

$$\begin{aligned}
 E_t &= E_p + E_k \\
 &= mgh + \frac{1}{2}mv^2 \\
 \Rightarrow \quad 2.925 \times 10^5 &= (2.25 \times 10^3)(9.80)(8.00) + \frac{1}{2}(2.25 \times 10^3)v^2 \\
 \Rightarrow \quad v &= 10.16 \text{ ms}^{-1}
 \end{aligned}$$

$\Sigma F = F_c = F_w + R$

If $R = 0$, then:

$$\begin{aligned}
 \Sigma F &= F_c = F_w \\
 \Rightarrow \quad \frac{mv^2}{r} &= mg \\
 \Rightarrow \quad v &= \sqrt{gr} \\
 &= \sqrt{(9.80)(4.00)} \\
 &= 6.261 \text{ ms}^{-1}
 \end{aligned}$$

i.e. The speed of the coaster is greater than the minimum speed required.

$\therefore$ The coaster makes it safely around the loop .

$$\begin{aligned}
 \text{(c) AT B: } \Sigma F &= F_c = F_w + R \\
 \Rightarrow R &= F_c - F_w \\
 &= \frac{mv^2}{r} - mg \\
 &= \frac{(2.25 \times 10^3)(10.16)^2}{(4.00)} - (2.25 \times 10^3)(9.80) \\
 &= 3.601 \times 10^4 \text{ N}
 \end{aligned}$$

$\therefore$ Force exerted = 3.60×10^4 N downwards.

Physics in action — How to travel upside down without falling out

You might have been on a roller-coaster like the one in Figure 1.26, where you were actually upside down at times during the ride. These rides use their speed and the radius of their circular path to prevent the riders from falling out. In theory, the safety harnesses worn by the riders are not needed to hold the people in their seats.



Figure 1.26

The thrill seekers on this roller-coaster ride do not fall out when upside down because the centripetal acceleration of the cart is greater than 9.8 m s^{-2} down.

The reason people do not fall out is that their centripetal acceleration while on the roller-coaster is greater than the acceleration due to gravity (-9.80 m s^{-2}). To understand the significance of this, try the following activity. Place an eraser on the palm of your hand, then turn your hand palm down and move it rapidly towards the floor. You should find, after one or two attempts, that it is possible to keep the eraser in contact with your hand as you 'push' it down. The eraser is upside down, but it is not falling out of your hand. Your hand must be moving down with an acceleration in excess of -9.80 m s^{-2} , and continually exerting a normal force on the eraser. This acceleration of 9.80 m s^{-2} down is the critical point in this exercise. If your hand had an acceleration less than this, the eraser would fall away from your hand to the floor. A similar principle holds with roller-coaster rides. The people on the ride do not fall out at the top because the motion of the roller-coaster gives them a centripetal acceleration that is greater than 9.80 m s^{-2} down. The engineers who designed the ride would have ensured that the roller-coaster moves with sufficient speed and in a circle of the appropriate radius so that this happens.

As an example, consider a ride of radius 15 m in a simple vertical circle (Figure 1.27). It is possible to calculate the speed that would ensure that a rider cannot fall out. We will assume that the person has a mass of 45 kg and that g is -9.80 m s^{-2} . At the critical speed, the reaction force of the seat, F_{seat} , on the person will be zero. In other words, the seat will exert no force on the person at this speed:

$$\begin{aligned}
 \uparrow + \\
 \downarrow -
 \end{aligned}
 \quad
 \begin{aligned}
 F_{\text{seat on person}} &= F_c + (-F_g) \\
 F_{\text{seat on person}} &= 0 \text{ N} \\
 \frac{mv^2}{r} &= mg \\
 r &= -15.0 \text{ m} \\
 g &= -9.80 \text{ m s}^{-2} \\
 v &= \sqrt{rg} \\
 &= \sqrt{(-15.0)(-9.80)} \\
 &= 12.1 \text{ m s}^{-1}
 \end{aligned}$$

This speed is equal to 43 km h^{-1} and is the minimum needed to prevent the riders from falling out. In practice, the roller-coaster would move with a speed much greater than this to ensure that there was a significant force between the patrons and their seats. Corkscrew roller-coasters can travel at up to 110 km h^{-1} and the riders can experience accelerations of up to 50 m s^{-2} ($5g$). So, safety harnesses are really only needed when the speed is below the critical value; their primary function is to prevent people from moving around while on the ride.

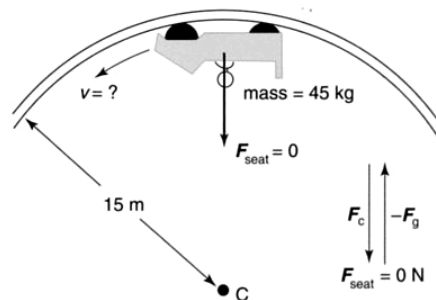


Figure 1.27

At the speed at which the reaction force of the seat on the person, F_{seat} , becomes zero, the centripetal force, F_c , on the rider will equal the force due to gravity, F_g , and they will move so that the centripetal acceleration is 9.80 m s^{-2} down.

Physics file

There are actually very few circular loop-the-loops used in roller-coasters. As there is a loss of kinetic energy as the coaster increases in height, there is a corresponding decrease in speed and therefore a decrease in the centripetal force. The centripetal force must remain large at the top so that we feel firm contact with the seat. Recall that centripetal force increases when the radius decreases, so if we can reduce the radius of the loop as the roller coaster travels up the loop, then the centripetal force can remain constant even though the speed is decreasing. On the way down, the radius increases so that the centripetal force is reduced as the coaster gains speed again. The loop that describes this shape is known as the clothoid loop.

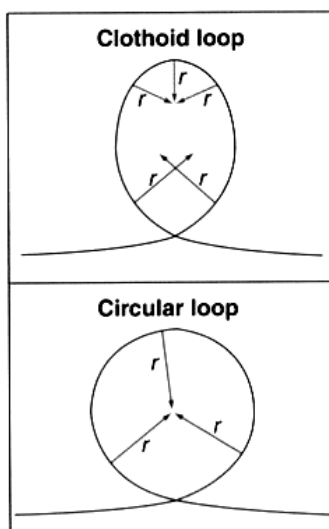


Figure fp.13 The changing radius of a clothoid loop enables a constant centripetal force despite a change in speed.

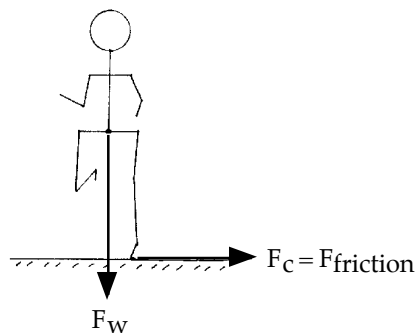
MOVING AROUND A CIRCULAR CURVE

Flat Curve

Consider a person running around a bend on an athletics track.

As seen in earlier notes, any object that moves around a horizontal curve must experience a centripetal force that is directed towards the centre of the curve.

If the person tries to run around the curve and **remain perfectly upright**, this **centripetal force is supplied only by the friction** between her shoe and the ground.

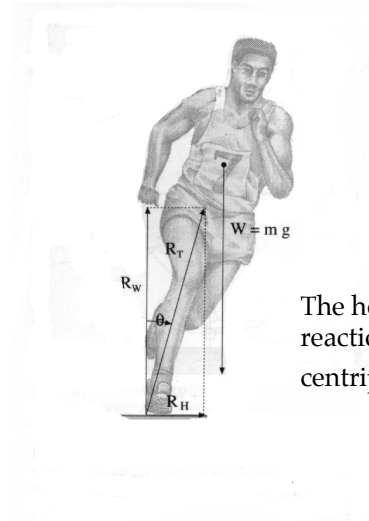


The person would experience a reaction force R directed vertically upwards from the ground.

The person will actually **lean into the curve**, making it easier to move around the curve.

By doing this, the **horizontal component of the reaction force** experienced by the person from the ground will be able to provide the centripetal force required.

(The force due to friction is much less than this value and is considered insignificant.)



The horizontal component of the reaction force (R_H) provides the centripetal force.

For calculations:

$$F_c = \vec{R} \cos \phi = \frac{mv^2}{r}$$

where ϕ = the angle between R and R_H .

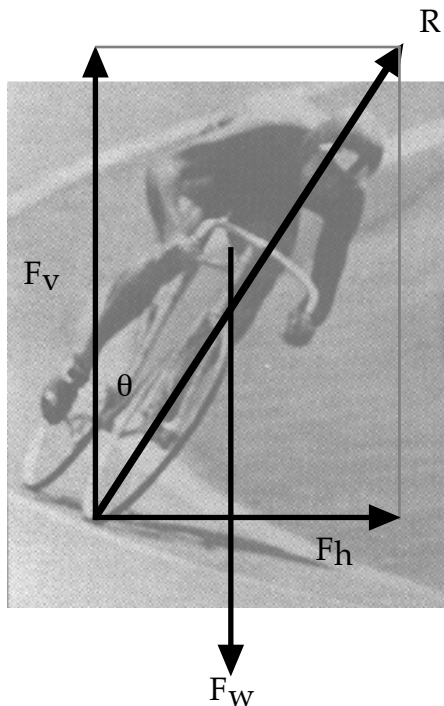
Banked Curves

In order for cars and trains to successfully negotiate corners at certain speeds, the road or track is banked at a particular angle.

This allows for the reaction force from the ground to exert a horizontal component that provides the centripetal force, thus reducing the reliance on friction alone.

Consider a cyclist riding around a velodrome with the *angle of banking* = θ .

If the frictional forces are considered negligible, then the following applies.



VERTICALLY

$$\begin{aligned}\Sigma F &= 0 \\ \Rightarrow F_v &= R \cos \theta = F_w \\ \Rightarrow R \cos \theta &= mg\end{aligned}\quad \text{Eqn 1}$$

HORIZONTALLY

$$\begin{aligned}F_h &= R \sin \theta = F_c \\ \Rightarrow R \sin \theta &= mv^2/r\end{aligned}\quad \text{Eqn 2}$$

Divide Eqn 2 by Eqn 1.

$$\begin{aligned}\Rightarrow R \sin \theta / R \cos \theta &= mv^2/r / mg \\ \Rightarrow \tan \theta &= v^2 / rg\end{aligned}$$

Note

This formula applies to *all* “banking” situations.

e.g. A plane or bird banking in flight.

A Frisbee or discus banking in flight.

The banking used on corners for roads and railways.

Try to remember how it is *derived* rather than rotely remembering it!

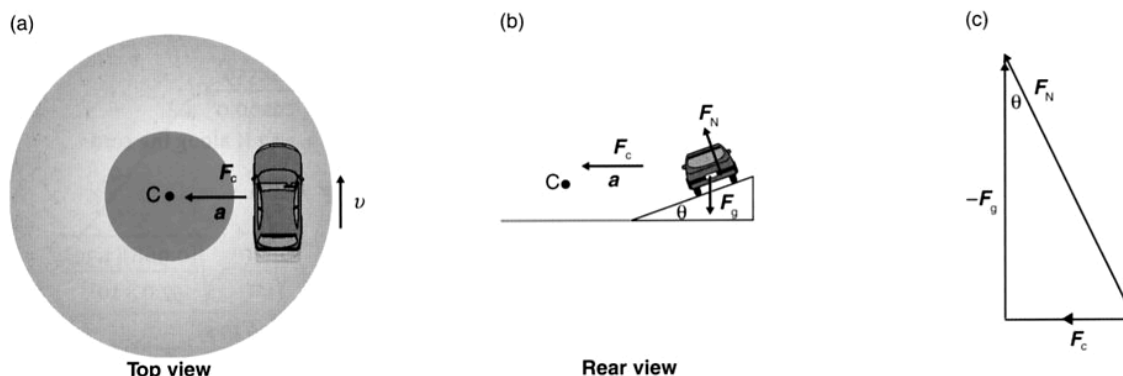


Figure 1.17

(a) The car is travelling in a circular path on a banked track. The centripetal acceleration and centripetal force are towards C. (b) The banked track means that the normal force has an inwards horizontal component. This is what enables the car to turn the corner. (c) Vector addition gives the centripetal force acting horizontally towards the centre.

Physics in action — Leaning into corners

In many sporting events, the participants need to travel around corners at high speeds. Motorbike riders lean their bikes over almost to the track as they corner. This leaning technique is also evident in ice skating, bicycle races, skiing and even when you run round a corner. It enables the competitor to corner at high speed without falling over. Why is this so?

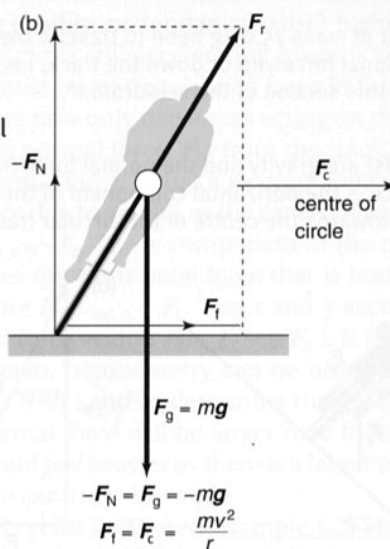
Consider a bike rider cornering on a horizontal road surface. The forces acting on the bike and rider (Figure 1.19b) are unbalanced. The forces are the force due to gravity, F_g , and the force from the track. The track exerts a reaction force, F_r , on the rider that acts both inwards and upwards. The inwards component is the frictional force, F_f , between the track and the tyres. The upwards component is the opposite of the weight force, $-F_g$, from the track.

The rider is travelling in a horizontal circular path at constant speed, and so has a centripetal acceleration directed towards the centre of the circle at C. Therefore, the centripetal force is directed towards C. By analysing the vertical and horizontal components in Figure 1.19b, we see that the force due to gravity, F_g , must balance the opposite of the weight force, $-F_g$. The centripetal force that is producing the centripetal acceleration is supplied by the frictional force, F_f . In other words, the rider is depending on a sideways frictional force to turn the corner. An icy or oily patch on the track would cause the tyres to slide out from under the rider, and he or she would slide painfully along the road at a tangent to the circular path.



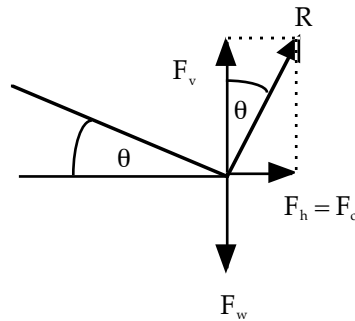
Figure 1.19

(a) Australia's Casey Stoner won the 2007 MotoGP championship. Here he is leaning his bike as he takes a corner at Phillip Island. Leaning into the corner enables him to corner at higher speeds. In fact, the bike would flip if he did not lean it. (b) The forces acting as the rider turns a corner are the force due to gravity, F_g , the opposite of the weight force, $-F_g$, and the friction, F_f , between the tyres and the road. The force due to gravity and the opposite of the weight force are balanced. The friction supplies the unbalanced centripetal force that leads to the circular motion.



Examples

- Westrail engineers have designed a curve on the rail line out to Kalgoorlie to allow the Prospector train to safely negotiate it at 90.0 kmh^{-1} . What must be the radius of the curve if the angle of the banking is 5.25° ?



$$\begin{aligned}
 \text{VERTICALLY} \quad \Sigma F &= 0 \\
 \Rightarrow F_v &= F_w \\
 \Rightarrow R \cos \theta &= mg \quad \text{Eqn 1}
 \end{aligned}$$

$$\begin{aligned}
 \text{HORIZONTALLY} \quad F_h &= F_c \\
 \Rightarrow R \sin \theta &= \frac{mv^2}{r} \quad \text{Eqn 2}
 \end{aligned}$$

$$\begin{aligned}
 \frac{\text{Eqn 2}}{\text{Eqn 1}} \Rightarrow \frac{R \sin \theta}{R \cos \theta} &= \frac{mv^2}{r} \times \frac{1}{mg} \\
 \Rightarrow \tan \theta &= \frac{v^2}{gr} \\
 \Rightarrow r &= \frac{v^2}{g \tan \theta} \\
 &= \frac{(25.0)^2}{(9.80)(\tan 5.25^\circ)} \\
 &= 6.941 \times 10^2 \text{ m}
 \end{aligned}$$

$$\therefore \quad \underline{\text{Radius of the curve} = 6.94 \times 10^2 \text{ m.}}$$

2.

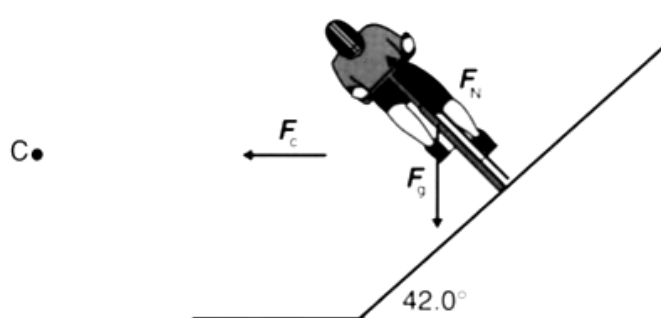
A curved section of track on an Olympic velodrome has radius of 50.0 m and is banked at an angle of 42.0° to the horizontal.

- Calculate the centripetal force acting on a cyclist riding at the design speed.
- At what speed would a cyclist of mass 75.0 kg need to travel if they were to experience zero frictional forces up or down the track, i.e. what is the design speed for this section of the velodrome?

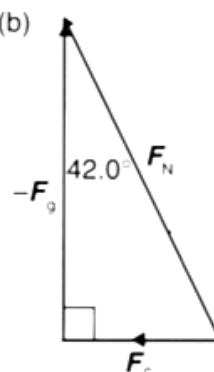
Solution

- The forces acting on the cyclist are gravity and the normal force from the track. The centripetal force is the horizontal component of the normal force and is directed towards the centre of the circular track as shown in diagram (a) below.

(a)



(b)



Adding the force vectors gives a right-angle triangle as shown in diagram (b). Trigonometry can be used to work out the net force.

$$m = 75.0 \text{ kg} \quad \tan \theta = \frac{F_c}{-F_g}$$

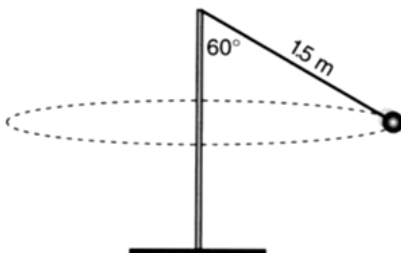
$$g = -9.80 \text{ m s}^{-1} \quad F_c = -mg \tan \theta = -(75.0)(-9.80)(\tan 42.0^\circ) \\ = 6.62 \times 10^2 \text{ N towards the centre of curvature}$$

- The centripetal force is used to find the speed:

$$m = 75.0 \text{ kg} \quad F_c = \frac{mv^2}{r} \\ r = 50.0 \text{ m} \quad v = \sqrt{\frac{(6.62 \times 10^2)(50.0)}{75.0}} \\ F_c = 6.62 \times 10^2 \text{ N} \quad = 21.0 \text{ m s}^{-1}$$

3. Conical Pendulum

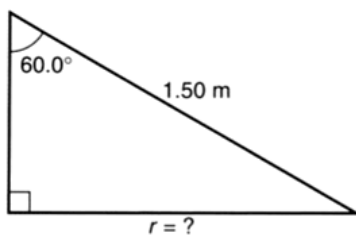
During a game of Totem-Tennis, the ball of mass 57.0 g is swinging freely in a horizontal circular path. The cord is 1.50 m long and is at an angle of 60.0° to the vertical shown in the diagram.



- Calculate the radius of the ball's circular path.
- Draw and identify the forces that are acting on the ball at the instant shown in the diagram.
- Determine the net force that is acting on the ball at this time.
- Calculate the size of the tensile force in the cord.
- How fast is the ball travelling at this time?

Solution

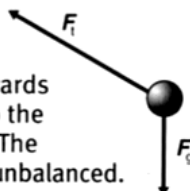
- a The centre of the circular path is not at the top end of the cord, but is at the point where the pole cuts the horizontal plane of the path of the ball. Trigonometry and a distance triangle can be used to work this out.



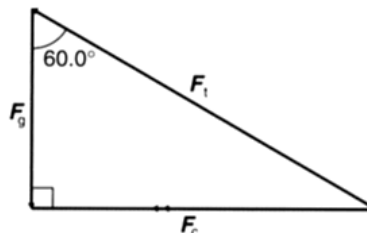
$$r = 1.50 \times \sin 60.0^\circ$$

$$r = 1.30 \text{ m}$$

- b There are two forces acting—the tension in the cord, F_t , and gravity, F_g . The vertical upwards component of the tension is always equal to the downwards force due to gravity on the ball. The horizontal component of the tension force is unbalanced.



- c The ball has an acceleration that is towards the centre of its circular path due to the unbalanced horizontal component of the tension force. This centripetal acceleration is horizontal and towards the centre of rotation. The centripetal force will also lie in this direction at this instant. A force triangle and trigonometry can be used here.



$$m = 57.0 \times 10^{-3} \text{ kg}$$

$$g = 9.80 \text{ m s}^{-2}$$

$$F_g = mg = (57.0 \times 10^{-3})(9.80)$$

$$= 0.559 \text{ N}$$

$$F_c = (0.559) \tan 60.0^\circ$$

$$= 0.968 \text{ N towards the centre}$$

- d Trigonometry can also be used to determine the tension in the cord.

$$F_g = 0.559 \text{ N} \quad \cos 60.0^\circ = \frac{F_g}{F_t}$$

$$F_t = \frac{0.559}{\cos 60.0^\circ}$$

$$= 1.12 \text{ N along the cord}$$

$$e \quad F_c = 0.968 \text{ N}$$

$$m = 57.0 \times 10^{-3} \text{ kg}$$

$$r = 1.30 \text{ m}$$

$$F_c = \frac{mv^2}{r}$$

$$v = \sqrt{\frac{F_c r}{m}} = \sqrt{\frac{(0.968)(1.30)}{57.0 \times 10^{-3}}}$$

$$= 4.70 \text{ m s}^{-1}$$

LAW OF UNIVERSAL GRAVITATION

Newton formulated this law as:

“The force between any two masses separated by a distance is an attractive force acting along the line joining the centre of the two masses.”

The gravitational force acting between two bodies, m_1 and m_2 :

- is one of attraction, and can be considered to act from the centre of each mass.
- acts equally and oppositely on each mass (Newton's third law): $F_1 = -|F_2$.
- is weaker if the masses are further apart. Gravitation acts in an inverse square manner.
i.e. $F \propto \frac{1}{r^2}$ where r is the distance in metres between the centres of the masses.
- depends directly on the mass of each body involved
i.e. $F \propto m_1$ and $F \propto m_2$.

Newton showed that the force acts as if the mass of each body is located at its centre of mass. This is why the distance of separation r must be measured from the centre of each object.

This is expressed as:

$$F = \frac{Gm_1m_2}{d^2}$$

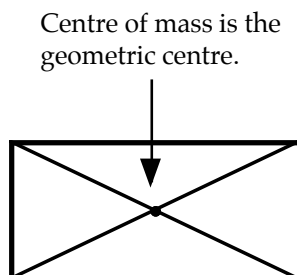
where F = the gravitational force (N)
 G = universal gravitational constant
 $= 6.67 \times 10^{-11} \text{ Nm}^2\text{kg}^{-2}$
 m_1, m_2 = the two masses (kg)
 d = distance between the centre of masses (m)

Note

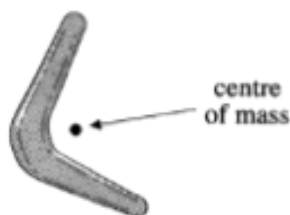
The **centre of mass** of an object is the point at which all the object's mass seems to act.

For a regular shaped object, the centre of mass is in the geometric centre.

e.g. a rectangular box.



For irregular objects such as boomerangs, the centre of mass is **outside** the body.



The simplest method to find the centre of mass of an object is to suspend it from two different points.

The point where two imaginary lines (showing the direction of the force due to gravity) cross, that is the centre of mass.

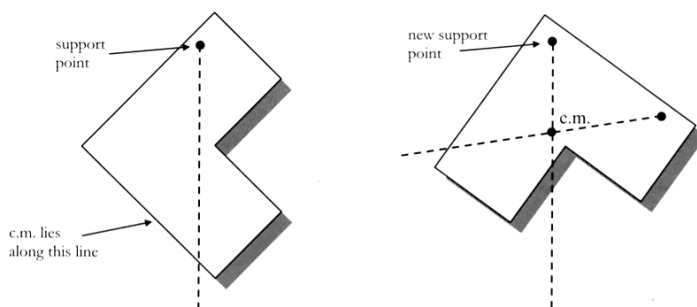


Figure 4.1 Finding centre of mass of a lamina

Terms that matter

- **Mass** Mass is the amount of matter in a body and is measured in kilograms. The mass of a body does not vary with position, e.g. a 2 kg hammer on earth will also have a 2 kg mass on the moon. A unit mass can be defined using $F = m a$.
- **Weight** Weight is a force and is measured in Newtons. Your weight is a measure of the gravitational attraction between you and the earth. It depends on where you are and the value of g at that point ($w = m g$). Your weight on the moon would be approximately that which it is on earth.
- **Centre of mass** For ease of calculation it is convenient to think of a single point in a body about which mass is evenly distributed. For regular shaped objects of uniform density this is their geometrical centre. It is also the point about which objects will smoothly rotate.
- **Centre of gravity** The centre of gravity of a body is the point at which all its weight may be considered to act. It is the point about which the body's weight is evenly distributed and in nearly all cases is the same as the centre of mass. Where bodies are very large or are in a non-uniform gravitational field, then the centre of gravity will be different to the centre of mass.

Example 1

The electron in a hydrogen atom is on average 5.29×10^{-11} m from the proton. Given that the mass of the electron is 9.11×10^{-31} kg and that of a proton is 1.67×10^{-27} kg, calculate the gravitational force between them.

Note The force due to gravity is insignificant when compared with the force due to the electrostatic attraction between the opposite charges.

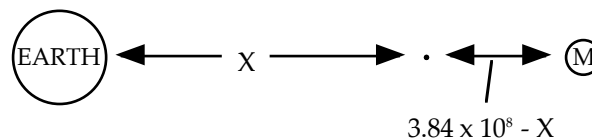
$$\begin{aligned} F &= \frac{Gm_1m_2}{d^2} \\ &= \frac{(6.67 \times 10^{-11})(1.67 \times 10^{-27})(9.11 \times 10^{-31})}{(5.29 \times 10^{-11})^2} \\ &= 3.626 \times 10^{-47} \text{ N} \end{aligned}$$

$$\therefore \quad \underline{\text{Force due to gravity} = 3.63 \times 10^{-47} \text{ N}}$$

Example 2

The Moon orbits about the centre of the Earth at a distance of 3.84×10^5 km. Where is the theoretical point between these two bodies where an object would experience no nett force of attraction due to them?

$$(m_{\text{Moon}} = 7.22 \times 10^{22} \text{ kg}, m_{\text{Earth}} = 5.98 \times 10^{24} \text{ kg})$$



At the theoretical point, the force due to both bodies are equal.

$$\begin{aligned} \text{i.e.} \quad F_{\text{Earth}} &= F_{\text{Moon}} \\ \Rightarrow \quad \frac{Gm_{\text{E}}m_{\text{object}}}{X^2} &= \frac{Gm_{\text{M}}m_{\text{object}}}{(3.84 \times 10^8 - X)^2} \\ \Rightarrow \quad \frac{(5.98 \times 10^{24})}{X^2} &= \frac{(7.22 \times 10^{22})}{(1.475 \times 10^{17} - 7.68 \times 10^8 X + X^2)} \\ \Rightarrow \quad (1.475 \times 10^{17} - 7.68 \times 10^8 X + X^2) &= 1.207 \times 10^{-2} X^2 \\ \Rightarrow \quad 0.9879 X^2 - 7.68 \times 10^8 X + 1.475 \times 10^{17} &= 0 \\ \Rightarrow \quad X &= \frac{7.68 \times 10^8 \pm \sqrt{(-7.68 \times 10^8)^2 - 4(0.9879)(1.475 \times 10^{17})}}{2(0.9879)} \\ &= 4.309 \times 10^8 \text{ m (too big) or } 3.465 \times 10^8 \text{ m.} \end{aligned}$$

$$\therefore \quad \underline{\text{Distance} = 3.46 \times 10^8 \text{ m from the Earth.}}$$

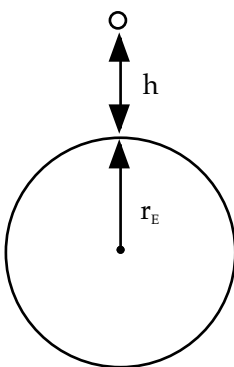
The gravitational field strength, or acceleration due to gravity, around an object is given by:

$$g = \frac{Gm}{d^2}$$

where g = acceleration due to gravity (ms^{-2})
 m = the mass of the object (kg)
 d = distance from the centre of mass of the object (m)

From this equation it follows that for a mass some height above the Earth's surface, the acceleration due to gravity is given by:

$$g = \frac{Gm_E}{d^2}$$



where m_E = the mass of the Earth (kg)
 d = the distance from the centre of the Earth (m)
 $= r_E + h$ (radius of Earth + height above ground)

Note

This formula can apply to *any object at any height above any heavenly body* such as the Moon, Jupiter, Mars, etc.

For a mass at the Earth's surface, the acceleration due to gravity is given by:

$$g = \frac{Gm_E}{r_E^2}$$

where r_E = the radius of the Earth (m)

Note

This equation can apply to all heavenly bodies as well.

e.g. Mars, Jupiter, the Moon, etc.

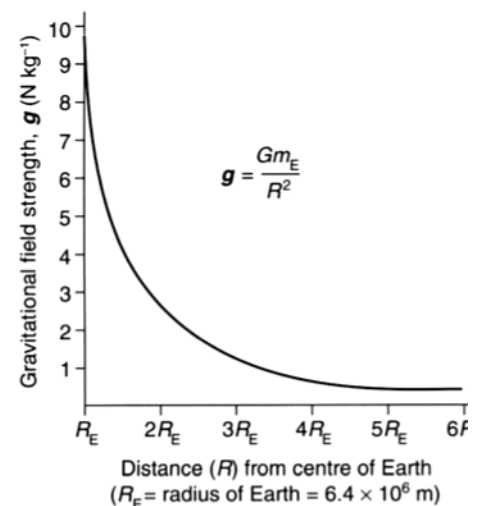
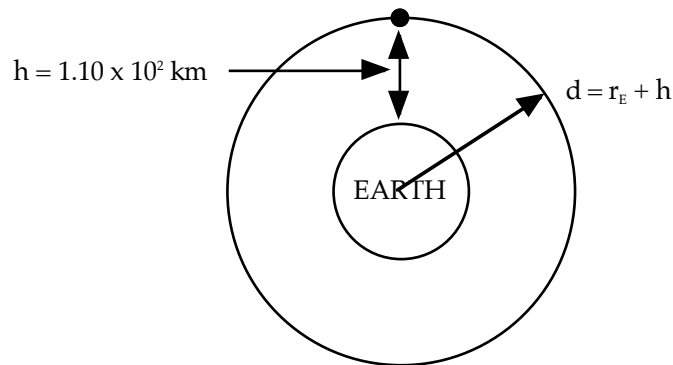


Figure 2.8

The gravitational field strength around the Earth. There is an inverse square relationship between the field strength, g , and the distance, R , from the centre

Example 1

The space shuttle orbits the Earth in a circular orbit at 1.10×10^2 km above the surface. Calculate the acceleration due to gravity at this height.



$$\begin{aligned} g &= \frac{Gm_E}{d^2} \\ &= \frac{(6.67 \times 10^{-11})(5.98 \times 10^{24})}{(6.38 \times 10^6 + 1.10 \times 10^5)^2} \\ &= 9.470 \text{ ms}^{-2} \end{aligned}$$

$\therefore$ Acceleration = 9.47 ms^{-2} towards Earth's centre.

Example 2

What is the acceleration on the Moon's surface, given its mass is 7.22×10^{22} kg and its radius is 1.74×10^6 m?

$$\begin{aligned} g_M &= \frac{Gm_M}{r_M^2} \\ &= \frac{(6.67 \times 10^{-11})(7.22 \times 10^{22})}{(1.74 \times 10^6)^2} \\ &= 1.591 \text{ ms}^{-2} \end{aligned}$$

$\therefore$ Acceleration = 1.59 ms^{-2} towards the centre.

Example 3

A 10.0 kg watermelon falls a short distance to the ground. If the Earth has a radius of 6.37×10^6 m and a mass of 5.98×10^{24} kg, calculate the:

- a gravitational force that the Earth exerts on the watermelon
- b gravitational force that the watermelon exerts on the Earth
- c acceleration of the watermelon towards the Earth
- d acceleration of the Earth towards the watermelon.

Solution

- a The gravitational force of attraction that the Earth exerts on the watermelon is:

$$\begin{aligned} m_{\text{wm}} &= 10.0 \text{ kg} & F_{\text{E on wm}} &= \frac{GMm}{r^2} \\ M_{\text{E}} &= 5.98 \times 10^{24} \text{ kg} & &= \frac{(6.67 \times 10^{-11})(5.98 \times 10^{24})(10.0)}{(6.37 \times 10^6)^2} \\ G &= 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2} & &= 98.3 \text{ N towards the Earth} \end{aligned}$$

- b The watermelon also exerts a force of attraction of 98.3 N towards the watermelon, on the Earth. These forces are an action/reaction pair. Even though the forces acting on the Earth and the watermelon are equal in size, the effect of these forces on each body is very different.

- c The acceleration of the watermelon is:

$$\begin{aligned} m_{\text{wm}} &= 10.0 \text{ kg} & F_{\text{E on wm}} &= ma_{\text{wm}} \\ F_{\text{E on wm}} &= 98.3 \text{ N} & a_{\text{wm}} &= \frac{98.3}{10.0} \\ & & &= 9.83 \text{ m s}^{-2} \text{ towards the Earth} \end{aligned}$$

This is, of course, very close to the average acceleration of all free-falling objects near the Earth's surface.

- d The acceleration of the Earth is:

$$\begin{aligned} m_{\text{wm}} &= 10.0 \text{ kg} & F_{\text{wm on E}} &= ma_{\text{E}} \\ F_{\text{wm on E}} &= 98.3 \text{ N} & a_{\text{E}} &= \frac{98.3}{5.98 \times 10^{24}} \\ & & &= 1.64 \times 10^{-23} \text{ m s}^{-2} \\ & & &\text{towards the watermelon} \end{aligned}$$

The motion of the Earth is hardly affected by the gravitational force of the watermelon.

Physics in action — How the mass of the Earth was first determined

The value of the universal gravitational constant, G , was first determined around 1844. The groundwork for this determination was done by Englishman Henry Cavendish in 1798. Cavendish, a rather eccentric man, was enormously wealthy and he devoted his life to scientific pursuits. He used the apparatus shown in Figure 2.4 to obtain the first direct measurement of the gravitational force of attraction between small objects. A large cage, Z, enclosed two pairs of lead spheres: a large pair, W, and a smaller pair, X, suspended from a metal rod that was itself suspended by a fine wire. The large spheres, initially close to the floor, were raised so that they were adjacent to but on opposite sides of the smaller spheres. The smaller spheres experienced a gravitational force of attraction towards the larger spheres. As they swung closer, the wire twisted. The amount of twist gave Cavendish a measure of the size of the gravitational forces. He needed a small telescope, T, illuminated by a lamp, L, to detect the tiny movement. These measurements enabled Cavendish to obtain the first calculation of the density of the Earth.

The mass of the Earth, M_E , could then be determined. The radius of the Earth, long known as 6.37×10^6 m, and the gravitational force on a 1.00 kg mass were used to calculate the Earth's mass:

$$\begin{aligned} F_g &= \frac{GM_E m}{r^2} \\ M_E &= \frac{F_g r^2}{Gm} \\ &= \frac{(9.80)(6.37 \times 10^6)^2}{(6.67 \times 10^{-11})(1.00)} \\ &= 5.96 \times 10^{24} \text{ kg} \end{aligned}$$

The value for G that was calculated later was only 1% different from the value that we use today.

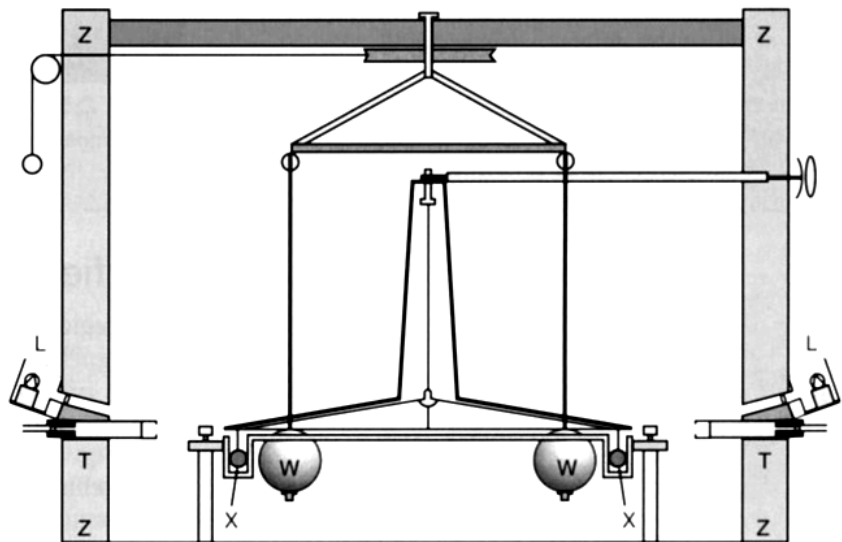


Figure 2.4

A diagram of the apparatus that was used by English physicist Henry Cavendish in 1798 to experimentally determine the value of the Earth's density

Physics in action — Gravity: the weakest force

Just four fundamental forces are responsible for the behaviour of matter, ranging from the smallest subatomic particles through to the most massive galaxies. These are the strong nuclear force, weak nuclear force, electromagnetic force and gravitational force.

The strongest of these forces, the **strong force**, is the force that binds the nucleus. It is a force of attraction that acts between nucleons (protons and neutrons), and is strong enough to overcome the repulsion between the protons in the nucleus. It only acts over a very short distance, of the order of 10^{-15} m, which is approximately the distance between adjacent nucleons.

The **weak force** is an extremely short-range force that acts inside atomic nuclei. It is responsible for radioactive processes such as beta decay. The weak force is much weaker than the strong nuclear and electromagnetic forces.

Electromagnetic forces act between charged particles: like charges repel and unlike charges attract. Compared with the strong force, the electromagnetic force is a long-range force. It acts to hold electrons in their orbits around atomic nuclei, but its strength varies as the inverse square of the distance between the charges. The electromagnetic force can act over an infinite distance.

The **gravitational force** is also a long-range force that acts between all bodies. Gravitational forces not only make things fall to the ground when they are dropped, they also extend across space and hold the planets in their orbits around the Sun—they hold galaxies together. The gravitational force affects the rate at which the Universe is expanding.

Physicists are currently scanning the Universe in an effort to find gravity waves. Some predict that collapsing stars will create ripples in space-time that we will be able to detect.

While gravitation, along with electromagnetism, has the longest range of the four fundamental forces, it is by far the weakest of these forces. Its strength is just 10^{-25} times that of even the weak force. Over extremely small distances, the strong, electromagnetic and weak forces are dominant and the gravitational attraction between the particles is of no significance. But on the much larger scale of outer space, the opposite situation arises: gravitation rules! The electromagnetic force has no influence over these distances because, at a distance, matter appears uncharged.



Figure 2.5

Gravitational forces hold planets in their orbits around the Sun.

Representing Gravitational Fields

A gravitational field, g , can be represented diagrammatically by lines with arrows that show the direction of the force on the object.

The gravitational field around the Earth is directed towards the centre of the Earth and is not uniform. It becomes weaker at higher altitudes.

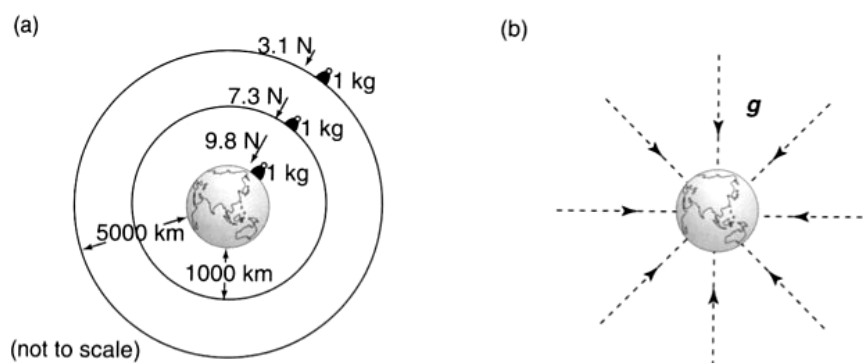


Figure 2.7

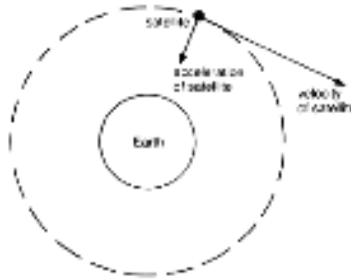
(a) The gravitational force acting on a 1.00 kg mass becomes smaller at greater distances from the Earth. (b) This non-uniform gravitational field, g , is represented by radial lines directed towards the centre of the Earth. The field is strongest where the lines are closest.

Motion Of Satellites

A satellite is any body that orbits about a larger (more massive) object in space.

e.g. The Moon about the Earth, the planets about the Sun, etc.

When a satellite is orbiting the Earth, a centripetal force is needed to maintain its circular orbit while a horizontal speed is required to keep it in the orbit.



The satellite is constantly accelerated by gravity towards the centre of the Earth while the horizontal speed maintains the orbit.

i.e. A gravitational force provides the centripetal force.

i.e.

$$\mathbf{F}_{\text{gravity}} = \mathbf{F}_{\text{centripetal}}$$

$$\Rightarrow \frac{Gm_E m_{\text{sat.}}}{r^2} = \frac{m_{\text{sat.}} v^2}{r}$$

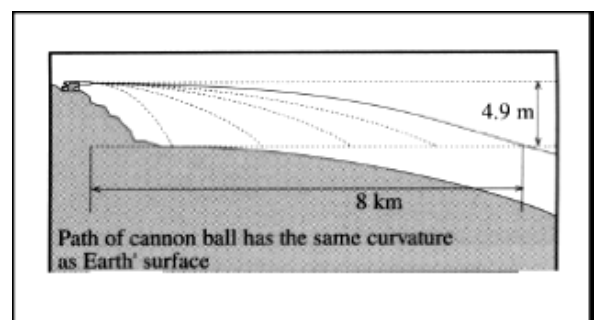
where

r = the radius of the orbit (m)
 $= r_{\text{Earth}} + h$ (height above the surface)

Consider the firing of a shell from a cannon horizontally from a height of 1.00 m above the ground.

It initially has no vertical motion, only a large horizontal velocity. It will immediately start to fall to the ground under the influence of gravity and will hit the ground.

The faster the shell is fired, the further it will travel horizontally before it hits the ground.



If the speed of the shell is increased to about 8.00 km s^{-1} , it will actually follow the curve of the Earth rather than hitting it.

i.e. It appears to be “falling around the Earth”.

Hence it has begun to orbit the Earth (although in the atmosphere friction would soon cause it to slow and fall to the ground).

Example 1

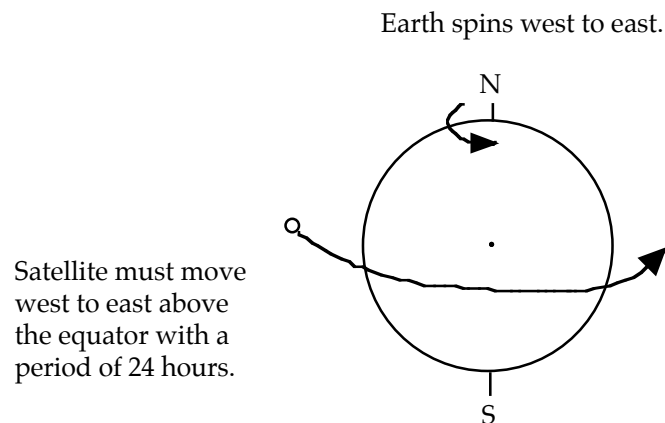
The Aussat satellites that provide television and communications for Australia are located in “geostationary” orbits - an orbit that keeps them over the same point on the Earth. They are located above the equator to the north of Australia.

- (a) Why must geostationary satellites be located above the equator and at no other latitude?
- (b) At which height above the Earth would these satellites be found?

- (a) Satellites must orbit west to east in a period of 24 hours to be above the same point on the Earth's surface.

Satellites must also orbit about the centre of mass of the Earth.

All of these conditions can only be met if the satellite is located directly above the equator.



$$\begin{aligned}
 (b) \quad & F_c = F_g \\
 \Rightarrow & \frac{mv^2}{r} = \frac{Gm_E m_s}{r^2} \\
 \Rightarrow & \frac{4\pi^2 r^2}{T^2} = \frac{Gm_E}{r} \quad (\text{since } v = \frac{2\pi r}{T}) \\
 \Rightarrow & r = \sqrt[3]{\frac{Gm_E T^2}{4\pi^2}} \\
 & = \sqrt[3]{\frac{(6.67 \times 10^{-11})(5.98 \times 10^{24})(24.0 \times 3.60 \times 10^3)}{4\pi^2}} \\
 & = 4.225 \times 10^7 \text{ m}
 \end{aligned}$$

Since $r = r_E + h$:

$$\begin{aligned}
 h &= r - r_E \\
 &= 4.225 \times 10^7 - 6.38 \times 10^6 \\
 &= 3.587 \times 10^7 \text{ m}
 \end{aligned}$$

$\therefore$ Satellite must be 3.59×10^7 m above the Earth's surface.

Example 2

Optus D2 is a geostationary satellite. Its period of orbit is 24.0 hours, so that it revolves at the same rate at which the Earth turns. Given that the mass of the Earth is 5.98×10^{24} kg and the mass of Optus D2 is 1160 kg, calculate:

- Optus D2's orbital radius
- the gravitational field strength at this radius
- Optus D2's orbital speed
- Optus D2's acceleration.

Solution

a $T = 24.0 \text{ h} = 8.64 \times 10^4 \text{ s}$

$$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$$

$$M_E = 5.98 \times 10^{24} \text{ kg}$$

$$F_c = F_g$$

$$\frac{m4\pi^2 r}{T^2} = \frac{GMm}{r^2}$$

$$r^3 = \frac{GMT^2}{4\pi^2}$$

$$r = \sqrt[3]{\frac{(6.67 \times 10^{-11})(5.98 \times 10^{24})(8.64 \times 10^4)^2}{4\pi^2}}$$

$$= 4.23 \times 10^7 \text{ m}$$

So the orbital radius of Optus D2 is 42 300 km. This is about 5.5 Earth radii from the surface of the Earth.

b $G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$

$$M_E = 5.98 \times 10^{24} \text{ kg}$$

$$r = 4.23 \times 10^7 \text{ m}$$

This satellite is in orbit at a great distance from the Earth and the gravitational field strength at its location is just 0.223 N kg^{-1} .

- c The speed of Optus D2 is given by:

$$T = 24.0 \text{ h} = 8.64 \times 10^4 \text{ s}$$

$$r = 4.23 \times 10^7 \text{ m}$$

$$v = \frac{2\pi r}{T}$$

$$= \frac{2\pi(4.23 \times 10^7)}{8.64 \times 10^4}$$

$$= 3.07 \times 10^3 \text{ m s}^{-1}$$

i.e. about 3 km s^{-1}

- d The acceleration of the satellite as it orbits is equal to the gravitational field strength at this radius as calculated in part b, i.e. $a = g = 0.223 \text{ m s}^{-2}$ towards the centre of the Earth.

Example 3

Ganymede is the largest of Jupiter's moons. It is about the same size as the planet Mercury. Ganymede has a mass of 1.66×10^{23} kg, an orbital radius of 1.07×10^9 m and an orbital period of 6.18×10^5 s.

- Use this information to determine the mass of Jupiter.
- Calculate the orbital speed of Ganymede.
- Calculate the gravitational force that Ganymede exerts on Jupiter.
- What is the size of the gravitational force that Jupiter exerts on Ganymede?



Solution

- a** To calculate the mass of Jupiter, use:

$$T = 6.18 \times 10^5 \text{ s}$$

$$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$$

$$r_G = 1.07 \times 10^9 \text{ m}$$

$$F_c = F_g$$

$$\frac{m_4 \pi^2 r}{T^2} = \frac{GMm}{r^2}$$

$$M = \frac{4\pi^2 r^3}{GT^2}$$

$$= \frac{4\pi^2 (1.07 \times 10^9)^3}{(6.67 \times 10^{-11})(6.18 \times 10^5)^2}$$

$$= 1.90 \times 10^{27} \text{ kg}$$

- b** The orbital speed of Ganymede is:

$$T = 6.18 \times 10^5 \text{ s}$$

$$r_G = 1.07 \times 10^9 \text{ m}$$

$$v = \frac{2\pi r}{T}$$

$$= \frac{2\pi(1.07 \times 10^9)}{6.18 \times 10^5}$$

$$= 1.09 \times 10^4 \text{ m s}^{-1}$$

i.e. about 11 km s^{-1}

- c** The gravitational force that Ganymede exerts on Jupiter is:

$$m_G = 1.66 \times 10^{23} \text{ kg} \quad F_g = \frac{GMm}{r^2}$$

$$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2} \quad F_g = \frac{(6.67 \times 10^{-11})(1.90 \times 10^{27})(1.66 \times 10^{23})}{(1.07 \times 10^9)^2}$$

$$r_G = 1.07 \times 10^9 \text{ m} \quad F_g = 1.84 \times 10^{22} \text{ N}$$

$$M_J = 1.90 \times 10^{27} \text{ kg} \quad F_{G \text{ on } J} = 1.84 \times 10^{22} \text{ N}$$

- d** The gravitational force that Jupiter exerts on Ganymede will also equal $1.84 \times 10^{22} \text{ N}$.

Physics in action — Kepler's laws

When Isaac Newton developed his law of universal gravitation, he was building on work previously done by Nicolaus Copernicus, Johannes Kepler and Galileo Galilei. Copernicus had proposed a Sun-centred (heliocentric) solar system, Galileo had developed laws relating to motion near the Earth's surface, and Kepler had devised rules concerned with the motion of the planets.

Kepler, a German astronomer, published his three laws on the motion of planets in 1609, about 80 years before Newton's law of universal gravitation was published. These laws are as follows.

- 1 The planets move in elliptical orbits with the Sun at one focus.
- 2 The line connecting a planet to the Sun sweeps out equal areas in equal intervals of time.
- 3 For every planet, the ratio of the cube of the average orbital radius to the square of the period of revolution is the same,

$$\text{i.e. } \frac{r^3}{T^2} = \text{constant.}$$

The most significant consequence of Kepler's laws was that planets were no longer considered to move in perfect circles at constant speeds. His first two laws proposed that planets moved in elliptical paths, and that the closer they were to the Sun, the faster they moved. It took Kepler many months of laborious calculations to arrive at his third law. Newton used Kepler's laws to justify the inverse square relationship. In fact, Kepler's third law can be deduced, for circular orbits, from Newton's law of universal gravitation:

$$\begin{aligned} F_g &= F_c \\ \frac{GMm}{r^2} &= \frac{4\pi^2 m}{T^2} \\ \frac{r^3}{T^2} &= \frac{GM}{4\pi^2} \end{aligned}$$

That is, for a given central body of mass M , the ratio $\frac{r^3}{T^2}$ is constant and equal to the collection of constants $\frac{GM}{4\pi^2}$ for all of its satellites. So, for example, if you know the orbital radius, r , and period, T , of one of the moons of Saturn, you could calculate $\frac{r^3}{T^2}$ and use this as a constant value for all of Saturn's moons. If you knew the period, T' , of a different satellite of Saturn, it would then be straightforward to calculate its orbital radius, r' .

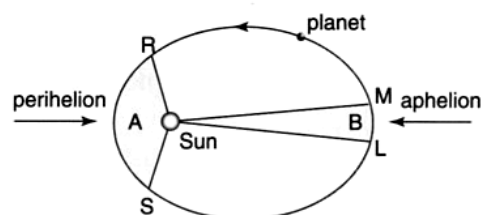


Figure 2.17

Planets orbit in elliptical paths with the Sun at one focus. Their speed varies continuously, and they are fastest when closest to the Sun. A line joining a planet to the Sun will sweep out equal areas in equal times, e.g. the time it takes to move from R to S is equal to the time it takes to move from L to M, and so area A is the same as area B.

Launching Satellites

Most satellites are launched near the equator.

Here the surface of the Earth has a spin speed of 1640 kmh^{-1} (464 ms^{-1}) which scientists use in order to give the satellites a horizontal speed. It significantly adds to the required speed of $8.00 \times 10^3 \text{ ms}^{-1}$ for a stable orbit.

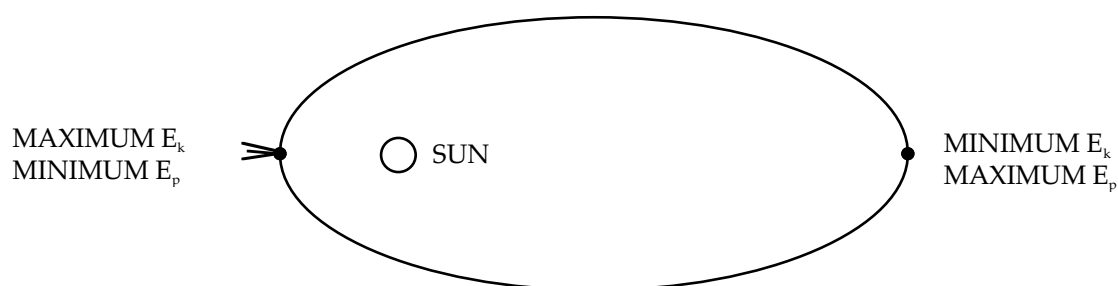
Satellites And Their Energy

Consider satellites such as comets that move in *elliptical orbits*. In fact most satellites that orbit the Earth have elliptical paths.

Such orbits mean that the satellite will vary its distance from the planet or Sun, resulting in different forces of gravitational attraction and hence orbital speeds.

Since such satellites neither gain nor lose energy (they aren't powered by rockets which add energy), decreasing the distance to the planet will decrease the E_p but increase the E_k (orbital speed).

i.e. $E_{\text{total}} = E_p + E_k$



If the energy of a satellite is changed, it will move to a higher or lower orbit depending on whether energy is added or removed.

e.g. **Speeding a satellite up will increase its E_k , causing it to move to a higher orbit.** As it does so, it will slow down to be stable in the new orbit. Thus the added energy will be seen as an increase in the E_p .

Slowing the satellite will remove energy from it. It will settle into a lower orbit where it will speed up. The change in energy is seen as a decrease in E_p but an increase in E_k .

Satellites that “decay” in orbit will lose energy, settle into the atmosphere and speed up, causing them to burn up due to the friction.